The War on Cash: Interoperability, Digital Payments, and Household Financial Fragility in Peru *

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Preliminary

Abstract

This paper estimates the causal effect of the Central Reserve Bank of Peru's (BCRP) Retail Payments Interoperability Strategy on payment instrument choice and household financial fragility in Peru. Interoperability connected previously segmented digital-wallet networks and payment rails, reducing compatibility and coordination frictions that limit the effective use of digital payments. I combine quarterly microdata from the National Household Survey (ENAHO, 2022–2024) with administrative information to build a predetermined district-level exposure measure based on the relative pre-rollout (December 2022) presence of institutions that become interoperable in each implementation phase. I exploit the staggered rollout and this territorial heterogeneity in a staggered-adoption difference-in-differences design following Callaway and Sant'Anna (2021), complemented with robustness checks and placebo tests. The results suggest a reallocation away from traditional instruments toward more digitized payment channels enabled by interoperability. In particular, I find a decline in cash use for everyday and recurring expenditures and a reduction in card use, alongside an increase in digital channel use (internet banking: +3.3 pp). In parallel, financial fragility falls by 2.3–2.4 pp, consistent with lower liquidity frictions and a greater ability to smooth shocks through timely transfers. The evidence is consistent with a technology channel, as more exposed districts experience a 7.5–7.9 pp increase in the probability of having prepaid mobile internet, in line with households acquiring the minimum connectivity needed to operate mobile payments. Effects concentrate in districts with higher pre-treatment connectivity, human capital, and formality, indicating that interoperability can accelerate the transition away from cash, although its reach remains constrained by persistent structural barriers.

JEL codes: E42, E41, D14, G21, O33, L86, D83.

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1 Introduction

How retail payments are organized shapes the cost of day-to-day transactions, the fluidity of commerce, and households' effective access to basic financial services. Yet even in economies where payment technologies diffuse rapidly, the transition toward digital instruments is not automatic. International evidence documents that cash often persists and that substitution away from it tends to be gradual (Bech et al., 2018; Von Kalckreuth et al., 2014; Rogoff, 2015). A natural reading is that the challenge is not purely technological but also one of market organization, because the adoption and use of digital payments face frictions in compatibility, coordination, and acceptance.

These frictions are especially salient in payment markets with network externalities and a two-sided structure. Consumers' incentives to adopt a payment instrument depend on its acceptance by merchants and other users, while merchants' investments in acceptance depend, in turn, on the expected mass of adopters. Recent empirical studies show that shocks that expand adoption on one side of the market can spill over to the other and shift the usage equilibrium (Higgins, 2024). In this framework, interoperability—understood as the integration of previously segmented networks—emerges as a natural policy to reduce compatibility and coordination costs, although its effects may depend on local frictions and the available infrastructure (Bianchi et al., 2023).

This paper studies the Retail Payments Interoperability Strategy promoted by the Central Reserve Bank of Peru (BCRP), which aimed to integrate previously segmented payment networks and digital wallets and to reduce frictions in digital transfers and payments (García et al., 2024). The policy was rolled out in phases and implemented in a context where cash use coexisted with a growing—yet geographically uneven—adoption of digital solutions. At the same time, connectivity and financial access exhibited persistent gaps. This setting provides meaningful variation to assess whether interoperability accelerates the transition toward digital payments and displaces traditional instruments such as cash and cards.

Empirically, the paper addresses four questions. First, it evaluates whether interoperability reduces cash use and how it reshapes its composition across expenditure categories. Second, it studies whether the policy affects the use of other instruments, in particular cards and digital channels. Third, it examines whether these changes are associated with improvements in outcomes related to financial inclusion and household welfare, proxied by household financial fragility (Lusardi et al., 2011). Fourth, it investigates whether connectivity operates as a mechanism that facilitates

adoption and use of digital rails.[1]

To identify causal effects, I exploit two sources of variation. The first comes from the staggered timing of the phase-based rollout. The second reflects predetermined cross-district heterogeneity in potential exposure to interoperability, proxied using pre-treatment information on the relative local presence of the institutions that subsequently become interoperable. Exposure is measured using district deposits and, in robustness exercises, augmented with measures of financial access infrastructure. Leveraging this variation, I estimate a staggered-adoption difference-in-differences design following Callaway and Sant'Anna (2021) and report robustness using alternative estimators (Sun and Abraham, 2021; De Chaisemartin and d'Haultfoeuille, 2020; Borusyak et al., 2024; de Chaisemartin and D'Haultfœuille, 2024), as well as placebo tests.

The results indicate economically meaningful shifts in payment choices. In more exposed districts, cash use declines in everyday and recurring categories. For instance, cash use for utilities falls by 4.4 percentage points (hereafter, pp) relative to a pre-treatment mean of 85.9%, while cash use for cooking fuels declines by 5.2 pp relative to a mean of 79.4%. These magnitudes correspond to reductions of roughly 5–7% from baseline. Card use also declines in several categories, such as utilities, where it falls by 1.7 pps relative to a pre-treatment mean of 3.1%. This pattern suggests that adjustment is not a simple "cash $\rightarrow$ card" substitution, but a shift toward interoperable account-to-account transfers and wallet-based payments with lower coordination and acceptance frictions. In parallel, the use of digital payment channels (internet banking and digital wallets) increases by 3.3 pp relative to a mean of 6.0%, and a broader indicator of digital-channel use rises by 5.0 pp, although the latter should be interpreted cautiously given questionnaire changes. On the welfare margin, point estimates suggest a 2.3–2.4 pp decline in a proxy for household financial fragility, about 11% of the pre-treatment mean. Because statistical support is marginal, I present this result as suggestive and consistent with reduced liquidity frictions and timelier risk-sharing.

The evidence is also consistent with a technological mechanism. In more exposed districts, access to prepaid mobile internet increases, a pattern consistent with households acquiring the minimum connectivity required to transact via mobile devices. Moreover, effects concentrate in districts with higher pre-treatment connectivity, human capital, and formality. This gradient points to complementarities: interoperability lowers compatibility and coordination frictions, but its translation into effective use depends on persistent local constraints.

[1] A key measurement limitation is that interoperable transactions are not directly observed at the individual level. I therefore infer substitution from changes in cash and card use, together with movements in proxies for the use of digital channels.

The paper contributes to two strands of the literature. First, it adds to the evidence on payment-technology adoption and network externalities by evaluating an intervention explicitly aimed at integrating networks and reducing compatibility frictions (Higgins, 2024; Bianchi et al., 2023). Second, it contributes to the literature on household financial inclusion and welfare by quantifying whether lowering frictions in digital payments is associated with changes in proxies for resilience and financial fragility (Jack and Suri, 2014; Riley, 2018; Lusardi et al., 2011).

2 Literature review

This paper sits at the intersection of four literatures: (i) the demand for cash and payment instrument choice; (ii) adoption of digital payments and network externalities in two-sided markets; (iii) financial inclusion, account ownership, and the use of formal financial services; and (iv) household financial well-being, especially resilience and fragility in the face of shocks. A common theme is that, although the diffusion of digital technologies has been rapid, cash often remains dominant and the effective use of digital payments faces coordination frictions and strong heterogeneity related to infrastructure and access (Bech et al., 2018; Von Kalckreuth et al., 2014; Rogoff, 2015).

2.1 Cash demand and the choice of payment instruments

A first strand studies why cash remains dominant even when electronic alternatives are available. On the theory side, Alvarez and Lippi (2009) shows that innovations that reduce the cost of accessing liquidity—such as more frequent and cheaper withdrawals—alter transaction demand for cash. In parallel, Rogoff (2015) emphasizes that cash provides services beyond transaction purposes, implying that displacing it may entail distributional and regulatory trade-offs that policy must weigh.

Micro evidence highlights that payment choice reflects systematic two-stage decisions. Individuals first adopt instruments (e.g., cards) and then choose the means of payment transaction by transaction based on relative costs, perceived attributes, and contextual features (Von Kalckreuth et al., 2014). This distinction between adoption and use is central for evaluating pro-digitalization policies, since greater technological availability does not necessarily translate into effective substitution away from cash. From a macro and sectoral perspective, Bech et al. (2018) documents that cash retains a large footprint in many countries, coexisting with electronic payments and generating patterns of partial substitution.

Taken together, this literature has two implications for the empirical analysis. First, policy can operate on distinct margins—adoption, use, and the composition of spending. Second, impacts on cash are likely to concentrate in transaction categories where digital payments are close substitutes and acceptance is relatively less costly. Building on these predictions, this paper estimates effects by expenditure category, a dimension that remains less explored in causal evidence on payment reforms in emerging economies.

2.2 Digital payment adoption, network externalities, and two-sided markets

The economics of payment systems is naturally framed through two-sided markets in which platforms intermediate interactions between consumers and merchants. With cross-side externalities, the structure of prices—not only their level—pins the equilibrium in adoption and use (Rochet and Tirole, 2003, 2006). This framework rationalizes, on the one hand, the tendency toward concentration driven by network effects and, on the other, the fragmentation that arises when competing networks are incompatible. In payments, compatibility and interoperability delimit the set of feasible transactions and the associated coordination costs, thereby shaping the speed of adoption and use. A synthesis of the two-sided approach and its empirical implications is provided by Rysman (2009).

On the empirical side, the most compelling causal evidence often exploits rollouts or plausibly exogenous shocks that push adoption on one side of the market and allow the researcher to identify spillovers to the other. Higgins (2024) uses the rollout of debit cards in Mexico and documents merchant acceptance responses and externalities affecting other consumers. Agarwal et al. (2024) exploits India's demonetization as a shock to cash reliance and finds a persistent increase in digital payment use, together with shifts in consumption spending. These designs illustrate the value of well-defined shocks for separating selection from causal effects.

Closer to this paper, an emerging literature discusses interoperability as a tool to reduce compatibility frictions and to mitigate the tension between concentration and fragmentation. Bianchi et al. (2023) structures the concept by distinguishing dimensions of interoperability and argues that its effects can be heterogeneous depending on local frictions and platform incentives. Empirically, Copestake et al. (2025) studies network integration in India and finds that interoperability increases total digital payment use, with larger effects where adoption was more fragmented ex ante. This evidence suggests that the marginal gains from interoperability depend on the initial ecosystem structure and motivates constructing a predetermined exposure measure based on the

relative presence of participants that become interoperable.

For Peru, additional evidence on the determinants of inclusion and digital adoption complements the two-sided framework. Sotomayor et al. (2018) shows that access to service points, education, and socioeconomic conditions are strongly associated with the use of formal services, while Robles et al. (2024) documents that digital skills, financial literacy, and trust predict the adoption of digital financial services even after controlling for observables. Taken together, these studies help interpret territorial heterogeneity and motivate why infrastructure and capabilities may condition effective use. However, their contribution is primarily descriptive and does not identify the causal effect of compatibility reforms between networks—such as interoperability—on instrument substitution and welfare.

Theory suggests that this emphasis on architecture is not incidental. Chiu and Wong (2022) shows that platforms' business-model choices and the design of the payment instrument can affect welfare, resilience, and incentives to interoperate, precisely because payment architecture shapes operational risk and the distribution of benefits. In that spirit, Andia et al. (2025) quantifies, using demand estimation with aggregate market-share data (2019–2024), which attributes drive the expansion of digital wallets in Peru. Their counterfactuals highlight the role of features such as immediate 24/7 payments, QR functionality, and compatibility with the acceptance infrastructure, and suggest positive associations with welfare among banked users. In parallel, evidence on interoperability in Peru has largely developed through institutional and implementation-oriented analyses (Ancalle and Garcia, 2024), as well as reports emphasizing the role of physical infrastructure—such as ATMs and banking agents—in inclusion and use (Chong and Ventura, 2024; Aurazo and Gasmi, 2024; Azevedo et al., 2025; FED, 2022). The common message is that compatibility and infrastructure matter, but the causal question remains open as to how much an interoperability reform displaces cash and card use, under what local conditions it does so, and whether it translates into measurable improvements in household financial well-being. This paper helps close that gap by leveraging quasi-experimental variation generated by a staggered rollout and a predetermined measure of territorial exposure.

In sum, the literature delivers a clear prediction: lowering compatibility frictions can shift the adoption-and-use equilibrium on both sides of the market. Yet causal evidence on interoperability remains scarce, especially when the objective is to quantify substitution away from traditional instruments rather than merely the expansion of digital payments. This paper contributes precisely by estimating that margin in a setting with a staggered rollout and predetermined heterogeneity

in exposure.

2.3 Digital payments and financial well-being

Another literature studies how digital payments connect to financial inclusion and access to credit. A key mechanism is that digital transaction histories reduce information frictions and can expand credit access, especially for new-to-credit individuals. Alok et al. (2024) analyzes public payment infrastructure and open banking in India and documents credit expansion by incumbents and fintechs, including in subprime segments. In a complementary vein, Carriere-Swallow et al. (2021) discusses the role of "stack"-type infrastructures as enablers of competition and service provision.

A second strand evaluates implications for consumption, shock smoothing, and resilience. In low- and middle-income countries, lowering transaction costs for transfers can expand support networks and improve risk-sharing. Jack and Suri (2014) shows that mobile money in Kenya helps insulate consumption from shocks, exploiting the expansion of the agent network as a source of variation. Riley (2018) finds evidence of consumption smoothing in response to climate shocks among mobile money users, while finding no significant spillovers to non-users within the same community. More recently, Agarwal et al. (2024) shows that an increase in digital payments can raise spending even after cash availability recovers, suggesting that digital payments may alter liquidity and behavioral frictions.

Finally, the financial-fragility literature emphasizes that a substantial share of households faces constraints in covering unexpected expenses or liquidity shocks, and survey-based proxies have become standard measures of resilience (Lusardi et al., 2011). For Peru, beyond institutional studies, there is micro evidence on usage barriers and trust. An experimental evaluation of agent networks and financial education in rural areas shows that reducing access costs and improving trust affects attitudes and account use, while also underscoring that access alone does not guarantee active use (Gertler et al., 2016). This evidence reinforces the idea of complementarities between infrastructure, trust, and effective use.

Overall, the literature suggests that payment innovations can generate welfare gains when they reduce transfer and liquidity frictions. Nonetheless, there is limited causal evidence explicitly linking a compatibility reform such as interoperability to changes in the instruments used in everyday transactions and to household-level measures of financial well-being. This paper addresses that gap by quantifying substitution across instruments and presenting evidence on a proxy for household financial fragility, while also exploring connectivity-related mechanisms.

In summary, the literature establishes that interoperability can reduce coordination frictions and expand digital payment use, but causal evidence on substitution away from traditional instruments—and on welfare effects—remains limited. Peru's staggered rollout of interoperability provides a unique opportunity to estimate these effects. The next section describes the institutional context and implementation design.

3 The Peruvian context: the Retail Payments Interoperability Strategy

By 2022, roughly 89% of Peruvians reported using cash to make payments, and 56% reported using only cash (Ipsos Perú, 2022). García et al. (2024) emphasize that this heavy reliance on cash reflects not only preferences for anonymity and operational simplicity, but also limited physical access to financial infrastructure (branches, ATMs, banking agents, and POS terminals) outside major cities, as well as perceived and actual costs of using digital instruments in low-income and highly informal settings (BCRP, 2023).

On the supply side of payment infrastructure, the country had been moving—albeit incompletely—toward a more digital ecosystem. The National Payment System comprised the real-time gross settlement system (LBTR) for large-value payments and the electronic clearinghouse (CCE) for retail interbank transfers, complemented by card payment agreements (APT) and electronic money payment agreements (APDE), such as BIM (BCRP, 2023). Within strictly retail payments, the most relevant digital instruments were: (i) debit and credit card payments at merchants; (ii) intra- and interbank transfers via internet banking, mobile banking, ATMs, and banking agents; and (iii) digital wallets linked to bank accounts (Yape, Plin) and to e-money (BIM) (García et al., 2024).

The BCRP's Digital Payments Indicator (IPD)[2] reports sustained growth in the number of digital transactions per capita between 2015 and 2022, rising from 29 to 174 digital payments per capita, with a pronounced acceleration starting with the COVID-19 shock in 2020. International comparisons, however, indicate that this level still lagged behind regional economies such as Brazil or Costa Rica, suggesting that digitalization was advancing but from a still-limited base and one heavily concentrated among urban, higher-income segments with better access to financial and telecommunications infrastructure (BCRP, 2023).

Looking at the composition of digital payments prior to interoperability, the ecosystem appears

[2]It includes client operations of LBTR participants and retail digital payments: credit and instant transfers via the CCE, intra- and interbank transfers via wallets and other digital channels, payment cards, direct debit, and e-money operations (BIM) (BCRP, 2025).

fragmented and strongly intrabank. In December 2022, intrabank transfers accounted for about 61% of the number and 79% of the value of non-cash retail payments, indicating that a large share of transactions were processed within each institution rather than over shared infrastructures such as the CCE (Vásquez, 2024). Digital wallets represented roughly 47% of retail payments by number, but only about 4% of total value, reflecting their predominant use for small-ticket transactions (García et al., 2024).

The main retail payment rails operated as "closed loops." Yape and Plin, which by 2022 concentrated most digital wallet users, enabled payments only within each ecosystem: a Yape user could not send funds directly to a Plin user, and vice versa.[3] Both wallets processed transactions over card infrastructure (Visa Direct) and did not interoperate with the CCE's instant transfers (García et al., 2024). Similarly, the card-payment rail operated separately from the account-to-account transfer rail, so that a user of instant transfers could not send funds from mobile banking to a digital-wallet user using the recipient's phone number or QR code (García et al., 2024). In the same vein, the APDE BIM functioned as a closed-loop e-money scheme without effective interconnection to bank accounts or bank-based wallets, fragmenting the user experience and limiting network externalities in digital payments, in addition to its limited reach and interconnection (Vega and Vásquez, 2022).

Fragmentation also manifested in the use of QR codes and in barriers to entry for new payment service providers. Before interoperability, QR codes issued by one provider (e.g., Yape) were not necessarily readable by other wallets, forcing some merchants to display multiple QR codes and raising adoption costs (García et al., 2024). In addition, payments fintechs faced obstacles to accessing bank accounts, the CCE, and transactional data, which constrained the supply of innovative solutions and reinforced concentration among a few incumbent actors (Vega and Vásquez, 2022; García et al., 2024).

3.1 The BCRP Interoperability Strategy

In response to the frictions described above, the BCRP introduced in 2022 a strategy designed to accelerate the widespread adoption of retail digital payments in Peru. The central objective was to enable any person to pay digitally regardless of the wallet, bank, or institution holding their funds. In economic terms, interoperability can be viewed as a tool to internalize network externalities: by connecting rails and payment agreements that were previously isolated, it expands each

[3] By end-2022, Yape had 11.8 million users, while Plin had 8.6 million users (Castillo et al., 2023).

provider's potential user base and reduces the value of continuing to rely on cash as the dominant payment instrument (Castillo et al., 2023; García et al., 2024).

Additional objectives included: (i) increasing convenience for end users by simplifying the payment experience and reducing time and monetary costs; (ii) improving payment-system efficiency by promoting the use of shared infrastructures such as the CCE for low-value payments; (iii) strengthening competition in the retail payments market by leveling the playing field between large and small institutions and including microfinance institutions and new providers; (iv) fostering financial inclusion by leveraging digital payments as an entry point into other formal financial services; and (v) laying the groundwork for further innovation in payment services and, eventually, for the international interconnection of payment systems (García et al., 2024).

In 2022, through Circular No. 024-2022-BCRP, the BCRP launched the Retail Payments Interoperability Strategy, implemented in four phases (see Figure A.1).[4]

3.1.1 Phase 1. Interoperability between Yape and Plin (March 2023)

Phase 1 centered on interoperability between the digital-wallet payment agreements managed by incumbent banks: the Yape payment agreement—led by Banco de Crédito del Perú (BCP)—and the Plin payment agreement—led by BBVA, Scotiabank, and Interbank (García et al., 2024). This phase required that Yape users be able to send and receive payments to and from Plin users, and vice versa, using either a phone number or a QR code, while keeping the underlying card infrastructure and payment agreements. In practice, this entailed interconnecting alias directories (phone numbers) and defining operational and settlement rules for P2P transactions across wallets, under BCRP supervision and with support from acquirers and technology processors.[5] The regulatory framework set March 31, 2023 as the deadline for interconnecting the main digital wallets (BCRP, 2022). Production began with a controlled group in the last week of March—including testing with the BCRP team—and the mass rollout to users was completed gradually between April 1 and May 8 (see Figure A.2 for further implementation details for Phase 1).

Early outcomes indicate that this phase quickly translated into a high volume of interoperable wallet-to-wallet transactions. By March 2024, interoperable transactions associated with Phase 1 exceeded 2.5 million operations per day, roughly 70 million transactions per month, with a clear

[4] Circular No. 0024-2022-BCRP, "Regulation on Interoperability of Payment Services provided by Payment Service Providers, Payment Agreements and Payment Systems," Central Reserve Bank of Peru (BCRP). Available at: https://www.bcrp.gob.pe/docs/Transparencia/Normas-Legales/Circulares/2022/circular-0024-2022-bcrp.pdf.

[5] A concise summary of the phases and the main actors is provided in García et al. (2025). Table A.1 reports the list of participating institutions by phase and period.

predominance of payments initiated from Plin to Yape (García et al., 2024). By March 2025, interoperable transactions reached 125 million (García et al., 2025), indicating that wallet interoperability had become a central rail for everyday P2P payments in Peru.

3.1.2 Phase 2. Instant transfers and interoperable QR codes (September 2023)

Phase 2 extended interoperability beyond wallets to include account-to-account instant transfers (TIN) and QR-code payments. This phase began in September 2023, enabling interoperability between digital wallets and instant-transfer services offered by banks, cajas, and finance companies through their mobile-banking apps using the user's phone number, and introducing interoperability among QR codes issued by different providers (Fernández and García, 2024).[6]

Phase 2 generalized the use of aliases for the Interbank Account Code (CCI), which are simplified identifiers—typically linked to the user's phone number—used in lieu of the full CCI in the transaction. This mechanism allows users to receive instant transfers from any institution without revealing full bank-account details (García et al., 2025). In addition, the policy leveraged the QR Code Registry and set technical parameters so that codes generated by acquirers and wallets could be read by any participating application, thereby reducing the need for multiple QR codes per merchant and lowering coordination costs for small businesses.[7]

Following Phase 2, instant transfers through the CCE expanded rapidly. Between December 2023 and December 2024, monthly operations rose from 1.5 to 18.2 million, and transacted value increased from S/ 0.2 to S/ 2.3 billion (García et al., 2025). In parallel, QR payments through acquirers (Niubiz, Izipay) grew alongside a decline in average ticket size, consistent with rising adoption for low-value purchases at physical merchants and small businesses (García et al., 2025).

[6] The Interoperability Strategy initially set June 30, 2023 as the deadline for entities offering instant payment functionalities, instant transfers, and QR-code payments to move interoperable services into production (BCRP, 2022). However, to ensure operational security and stability during implementation and to formally incorporate technology providers into the scheme, the BCRP Board modified these deadlines through Circular No. 0013-2023-BCRP, extending the maximum go-live date to September 13, 2023, and to December 31, 2023 for entities that did not yet offer instant transfers (Circular No. 0013-2023-BCRP, "Amends Circular No. 0024-2022-BCRP approving the Interoperability Strategy of Payment Services provided by Payment Service Providers, Payment Agreements and Payment Systems," BCRP. Available at: https://www.bcrp.gob.pe/docs/Transparencia/Normas-Legales/Circulares/2023/circular-0013-2023-bcrp.pdf. See also the annex: https://www.bcrp.gob.pe/docs/Transparencia/Normas-Legales/Circulares/2023/circular-0013-2023-bcrp-anexo.pdf.).

[7] The Registry of QR Code and Digital Wallet Providers (Registro QR) was established by the Regulation on QR-code payment services, approved through Circular No. 0003-2020-BCRP. Available at: https://www.bcrp.gob.pe/docs/Transparencia/Normas-Legales/Circulares/2020/circular-0003-2020-bcrp.pdf.

3.1.3 Phases 3 and 4. Integrating e-money (December 2023) and open payments

Phase 3 sought to incorporate e-money issuers into the interoperable scheme, in particular the BIM payment agreement, by enabling direct or sponsored access to the CCE and the LBTR (Vásquez, 2025). Compartamos Banco played a central role as BIM's sponsoring institution in connecting it to the interoperable infrastructure, while other e-money issuers have gradually adopted the new scheme. For this study, I do not construct a phase-specific district-level exposure measure for Phase 3, as I lack consistent information to assign additional territorial variation relative to earlier phases. Moreover, available evidence suggests that its relative scale has been limited. As of June 2025, BCRP reports attribute 70.7% of interoperable transactions to Phase 1, 27.6% to Phase 2, and 1.7% to Phase 3 (García et al., 2025). This composition is consistent with Phase 3 not yet being a quantitatively important driver of interoperability, and the analysis therefore focuses on the effects of Phases 1 and 2.

Finally, Phase 4 envisages an *open payments* framework based on payment initiators and standardized APIs, with the goal of extending interoperability to new providers—such as fintechs, bigtechs, or telecommunications operators—under user consent. This stage remains under regulatory and technical design and is outside the analysis period (García et al., 2025).

4 Data and descriptive statistics

4.1 National Household Survey (ENAHO)

The main data source is the National Household Survey on Living Conditions and Poverty (ENAHO) conducted by Peru's National Institute of Statistics and Informatics (INEI). ENAHO is nationally representative, has been fielded since 2003, and covers both urban and rural areas in all departments and in Callao. For our analysis, I use quarterly microdata for the 2022–2024 period[8], which allow the construction of comparable individual- and household-level indicators of payment-method use and socioeconomic conditions.[9]

Following Aurazo and Vega (2021), I exploit ENAHO's employment module, which since 2015 has asked adults (18 years or older) about financial inclusion and the instrument used to pay for

[8] The baseline analysis excludes pre-2022 data to avoid introducing the COVID-19 pandemic as a confounding factor. For the placebo analysis, I use data from 2021Q3 through 2023Q1 to test for the absence of spurious effects around the onset of treatment while keeping pandemic exposure limited (see Section 8 for details).

[9] According to INEI technical documentation, ENAHO relies on a probabilistic, stratified, multi-stage sample design. It supports annual inference domains and, at least for national, urban, and rural aggregates, quarterly inference.

purchases in specific categories of goods and services. In particular, the questionnaire records the payment method used (e.g., cash, debit card, credit card, and mobile/internet banking, among others) for nine consumption categories, capturing primarily P2B (person-to-business) decisions rather than P2P (person-to-person) transactions. Consistent with this structure, I construct binary indicators of cash use and card use by expenditure category (e.g., food, utilities, fuels, clothing, etc.), as well as aggregate measures of internet-banking/digital-payment use.[10] Regarding internet-banking use, INEI changed the wording of the question in 2024 by explicitly incorporating additional channels (mobile banking and ATMs), unlike earlier years when the item referred exclusively to internet banking. To preserve intertemporal comparability for the Internet Banking indicator, I restrict the sample to 2022–2023 when estimating this outcome. As a complementary exercise, I also construct an aggregate (proxy) indicator of digital payments that combines internet-banking use with the use of digital wallets.[11] I also define an individual-level measure of financial inclusion equal to one if the respondent reports having an account and/or a card (debit or credit), and zero otherwise.

To proxy short-run liquidity vulnerability and financial stress—a potentially relevant mechanism in the transition from cash to digital instruments in contexts of volatile income—I use ENAHO's module on governance, democracy, and transparency. This module asks about the household's current economic situation (whether it is able to save, whether it barely balances income and expenditures, or whether it is forced to draw down savings or to borrow). Based on this question, I construct a financial-fragility dummy equal to one if the respondent reports that the household currently *(i)* must spend down savings or *(ii)* must borrow, and zero otherwise.[12]

In addition, because digital infrastructure is a key determinant of electronic-payment adoption, the baseline analysis incorporates measures of internet access: fixed internet and mobile internet (prepaid and postpaid). As a complement, I construct a set of connectivity variables used exclusively to explore effect heterogeneity across districts' initial (pre-interoperability) conditions, including internet service use, mobile phone ownership, and having one's own phone, among

[10] Respondents can report more than one instrument for a given category. The survey does not provide intensity (number of payments) nor transaction amounts, so these variables capture extensive-margin adoption/use.

[11] In practice, this proxy is a dummy equal to one if the individual reports using internet banking and/or digital wallets, and zero otherwise. Its interpretation is more limited than the main measure because, since comparable internet-banking information is restricted to 2023, wallet adoption dynamics in earlier periods may be partially absorbed by the internet-banking variable. This could bias estimates upward if both channels respond to different mechanisms or display misaligned adoption patterns. Given the absence of a consistent question covering the full set of digital channels over the entire period, I report this outcome only as complementary evidence and interpret it cautiously.

[12] This construction captures an operational margin closely related to liquidity constraints: the need to dissave/borrow to sustain current expenditures.

others (see Chapter 9 for details).[13]

Finally, I include a broad set of socioeconomic covariates constructed from standard ENAHO modules[14]: (i) education (primary, secondary, and higher); (ii) monetary poverty; and (iii) labor informality and receipt of social program transfers (Juntos and Pensión 65).[15]

4.2 Superintendency of Banking, Insurance and Private Pension Funds (SBS)

4.2.1 Deposits by branch and institution

I complement ENAHO with administrative data from Peru's financial supervisor (SBS). Specifically, I use the reports on Direct Credit and Deposits by Branch, available in the Financial System Statistics section of the SBS website.[16] These reports provide balances at the branch and institution level, allowing me to observe simultaneously (i) the local mass of formal savings mobilized in each territory and (ii) its composition by institution, an input that is crucial to proxy each district's exposure to the interoperability strategy.

For the empirical design, I use only two cross-sections (December 2022 and March 2023) of total deposits.[17] The December 2022 cross-section is the main *pre-treatment* measurement because it precedes the interoperability rollout and supports constructing a predetermined exposure measure based on the relative local presence of participating institutions in each phase. The March 2023 cross-section is used as a robustness exercise to assess the sensitivity of results to an alternative exposure measurement taken close to the start of the rollout.

4.2.2 ATMs and banking agents

To proxy local financial infrastructure beyond the presence of branches, I use SBS information on the number of automated teller machines (ATMs) and banking agents/correspondents by

[13] This approach allows testing whether the impact of interoperability differs systematically between districts with higher versus lower pre-existing digital penetration, consistent with the idea that new-technology adoption may depend on access constraints and on reaching a critical mass of users.

[14] I primarily rely on the education, summary, and employment and income modules.

[15] As with the connectivity variables, these socioeconomic characteristics are used primarily to examine heterogeneity across districts' initial conditions. For instance, I study whether interoperability generated larger impacts in districts with higher human capital, lower poverty, or lower informality prior to implementation, or whether effects were relatively larger in more constrained territories.

[16] These series are constructed from information submitted by supervised institutions through annexes and reports from the Accounting Manual. See https://www.sbs.gob.pe/normativa-y-estandares/normativa/normativa-sbs/plan-de-cuentas/planes-de-cuenta-del-sistema-financiero/informacion-complementaria-de-manual-de-contabilidad. The statistical information is available at https://www.sbs.gob.pe/estadisticas-y-publicaciones/estadisticas-/sistema-financiero.

[17] Under SBS definitions, *total deposits* correspond to the sum of demand, savings, time, and CTS deposits. Monetary variables are reported in thousands of soles, converting foreign currency at the month-end accounting exchange rate.

Table 1: Descriptive statistics

Variable	Mean	SD	Min	Max	N
Panel A: Cash use					
Groceries	0.92	0.28	0	1	233,196
Ready-to-eat food	0.91	0.28	0	1	233,196
Laundry	0.90	0.30	0	1	233,196
Utilities	0.86	0.35	0	1	233,196
Cooking fuels	0.79	0.41	0	1	233,196
Personal hygiene	0.91	0.28	0	1	233,196
Clothing and footwear	0.85	0.36	0	1	233,196
Furniture and appliances	0.29	0.46	0	1	233,196
Household appliances	0.11	0.31	0	1	233,196
Other	0.87	0.34	0	1	233,196
Panel B: Card use					
Groceries	0.04	0.20	0	1	233,196
Ready-to-eat food	0.04	0.19	0	1	233,196
Laundry	0.04	0.19	0	1	233,196
Utilities	0.02	0.15	0	1	233,196
Cooking fuels	0.01	0.11	0	1	233,196
Personal hygiene	0.04	0.18	0	1	233,196
Clothing and footwear	0.04	0.20	0	1	233,196
Furniture and appliances	0.01	0.10	0	1	233,196
Household appliances	0.01	0.10	0	1	233,196
Other	0.02	0.14	0	1	233,196
Panel C: Financial inclusion, fragility, internet banking, and digital payments					
Financial inclusion	0.52	0.50	0	1	233,196
Financial fragility	0.19	0.39	0	1	95,047
Internet banking use	0.06	0.25	0	1	155,995
Internet banking and digital payments	0.25	0.43	0	1	77,201
Panel D: Internet connectivity and mobile phone					
Fixed internet connection	0.30	0.46	0	1	142,380
Postpaid mobile internet connection	0.52	0.50	0	1	141,268
Prepaid mobile internet connection	0.68	0.47	0	1	141,268
Internet use	0.68	0.47	0	1	233,126
Mobile phone access	0.96	0.21	0	1	233,215
Own mobile phone	0.83	0.38	0	1	233,127
Panel E: Socioeconomic variables					
Age 18–24	0.15	0.35	0	1	233,196
Age 25–40	0.29	0.45	0	1	233,196
Age 41–64	0.39	0.49	0	1	233,196
Age 65+	0.17	0.38	0	1	233,196
Elementary education	0.31	0.46	0	1	233,127
High school education	0.38	0.48	0	1	233,127
Technical education (non-university)	0.14	0.35	0	1	233,127
University education	0.17	0.38	0	1	233,127
Poverty	0.22	0.41	0	1	233,215
Informal employment	0.78	0.42	0	1	174,417
Urban	0.68	0.47	0	1	233,196
Female	0.52	0.50	0	1	233,196
Juntos transfer income	0.05	0.22	0	1	233,196
Pension 65 transfer income	0.05	0.22	0	1	233,196

Note: The descriptive statistics correspond to the 2022–2024 period and are consistent with the figures reported by Montero and Delgado (2023). All variables in the table are dummies that equal one if the condition holds and zero otherwise.

institution–district.[18]

As part of the empirical analysis, I build this dataset for December 2022 to construct an alternative exposure measure based on financial-access infrastructure (ATMs and banking agents).[19] This measure is used as a robustness exercise, comparing results against the deposit-based exposure measure by branch–district. Incorporating these complementary channels helps, on the one hand, capture the presence of financial services in districts without formal branches but with in-person access points that facilitate basic operations (withdrawals, deposits, and low-value transactions). On the other hand, it leverages institution-by-district granularity to more precisely measure the relative presence of participating institutions, reducing potential measurement error in exposure.

5 Identification strategy

Given the implementation design of Peru's retail payments interoperability strategy—rolled out in a staggered fashion across phases and progressively incorporating different sets of participants—the analysis exploits variation in treatment timing and in potential exposure intensity at the district level. In a market where adoption and effective use of digital payment instruments depend, in part, on the pre-existing local presence of financial institutions and their associated infrastructure, it is natural to proxy treatment through a predetermined exposure measure that captures, in each district, the scope for interaction between the institutions that become connected in each phase.

Identification relies on a staggered difference-in-differences design that compares the evolution of outcomes between exposed and non-exposed districts around each implementation milestone, under the assumption of parallel trends in the absence of interoperability. The baseline estimation follows Callaway and Sant'Anna (2021), which identifies cohort- and relative-time effects and aggregates them transparently, avoiding the biases of TWFE in staggered settings with heterogeneous effects (De Chaisemartin and d'Haultfoeuille, 2020; Sun and Abraham, 2021). As

[18] In SBS regulation, these service points are classified as *complementary channels of customer service* distinct from branches, including (i) banking agents, (ii) basic-operations establishments, and (iii) ATMs. For expositional convenience, I use "banking agents" as a synonym for "correspondent agents." For the analysis, I include only the number of ATMs and banking agents. SBS produces this information from regulatory reports submitted by supervised institutions. Report No. 30 of the SBS Accounting Manual records, among other fields, the UBIGEO (department–province–district), address and geographic coordinates of the service point, as well as the total number of ATMs by address/network and the number of banking agents by establishment. The report indicates that the information corresponds to channels available at the end of each quarter (March, June, September, and December), and instructs institutions to report ATMs even when they operate through third-party networks under agreement, as well as banking agents affiliated directly or through aggregators.

[19] SBS updated the applicable format of Report No. 30 starting with the information corresponding to December 2022, which is consistent with the baseline period used in the analysis.

robustness, I replicate the results using alternative estimators proposed by Sun and Abraham (2021), Borusyak et al. (2024), and de Chaisemartin and D'Haultfœuille (2024), verifying that the main conclusions do not hinge on a single aggregation scheme or on restrictive assumptions about heterogeneity and the composition of the control group.

5.1 Exposure measure

Given the Peruvian institutional setting, interoperability reduced transaction frictions between payment networks that had previously been segmented and that required users either to coordinate across multiple apps/rails and/or to hold balances in more than one wallet. Accordingly, economically relevant district-level exposure does not depend solely on the size of a single block—i.e., the set of participants or financial institutions that belonged to distinct networks prior to the strategy—but rather on the potential for cross-network matches between two blocks of institutions that become interoperable in a given phase.

For example, in Phase 1, interoperability connected two pre-existing ecosystems—*Yape* on one side and *Plin* on the other—so that the marginal benefit of the policy in a district does not depend only on how widespread a single wallet was (e.g., *Yape* alone), but on the degree to which both networks coexisted locally.

I consider a district to be exposed when, prior to interoperability, comparable masses of deposits were present in institutions belonging to both blocks. In the baseline (preferred) exposure measure, this coexistence is proxied using the district distribution of total pre-treatment deposits—measured at τ = December 2022—for the financial institutions that comprise each block in each phase of implementation.[20]

I use administrative information on deposits at the district–institution–month level. Using pre-treatment deposits offers three advantages: (i) it anchors a predetermined measure and mitigates simultaneity concerns (deposits are not affected by interoperability at τ); (ii) deposits reflect the local stock of formal financial relationships and therefore the potential mass of users who could benefit from interoperable transactions; and (iii) it allows construction of exposure with full cov-

[20] As a robustness exercise, I construct an alternative exposure measure that is not limited to pre-treatment deposit mass but also incorporates the operational presence of participating institutions in each block. Specifically, in addition to deposits, I use two proxies of district-level financial-access infrastructure: (i) the number of ATMs and (ii) the number of banking agents associated with the same institutions. This measure aims to capture not only the pre-existing financial mass but also the operational capillarity through which users can fund, withdraw, and transact using financial services linked to participants in each block. Results using this alternative measure, along with other robustness checks, are reported in Section 8.

erage and homogeneous measurement across districts.

Given the staggered nature of treatment, for each phase $k \in \{\text{F1}, \text{F2}, \text{BN}\}$ there are two sets of institutions G_{1k} and G_{2k} (blocks) that become interoperable.[21] Let $\text{Dep}_{db\tau}$ denote the amount of deposits in district d at institution b, measured at τ. I define the deposit mass in district d associated with block G in phase k as[22]:

$$D_{dk}^{G}(\tau) = \sum_{b \in G} w_{b,G} \, \text{Dep}_{db\tau}. \tag{1}$$

The relative share of block G_{1k} in district d is defined as:

$$s_{dk}(\tau) = \frac{D_{dk}^{G_{1k}}(\tau)}{D_{dk}^{G_{1k}}(\tau) + D_{dk}^{G_{2k}}(\tau)}. \tag{2}$$

Based on $s_{dk}(\tau)$, I construct the phase-specific exposure index as:

$$E_{dk}(\tau) = \frac{2\, s_{dk}(\tau)\, \left[1 - s_{dk}(\tau)\right]}{0.5} = 4\, s_{dk}(\tau)\, \left[1 - s_{dk}(\tau)\right] \; \in [0, 1]. \tag{3}$$

Intuitively, $2s(1-s)$ is proportional to the probability that a district is exposed to interoperability—the normalization by 0.5 ensures a maximum of 1. $E_{dk}(\tau)$ is maximized when both blocks are balanced ($s_{dk} \approx 0.5$), and approaches zero when one block dominates ($s_{dk} \approx 0$ or 1). In districts where $D_{dk}^{G_{1k}}(\tau) + D_{dk}^{G_{2k}}(\tau) = 0$, I set $E_{dk}(\tau) = 0$, interpreting this as no economically meaningful exposure.[23] Appendix A provides additional details on the construction of the exposure measure. Figure A.3 displays the distribution of exposure, and Panel (a) of Figure A.4 shows its spatial distribution by phase.

5.2 Main empirical strategy

Exposure-induced treatment cohorts. Let t denote calendar quarter and d denote district. I define districts as "exposed" in phase k if $E_{dk}(\tau) > 0$. The first treatment date T_d is assigned as the

[21] For a detailed list of participating institutions by phase, see Table A.1.

[22] To allow for institutions that belong to both blocks (overlap), I assign weights $w_{b,G} \in [0, 1]$. In particular, if an institution belongs simultaneously to G_{1k} and G_{2k}, I weight it 50/50 in both blocks so that its total mass is not double-counted:

$$w_{b,G_{1k}} = w_{b,G_{2k}} = \tfrac{1}{2} \quad \text{if } b \in G_{1k} \cap G_{2k}, \qquad w_{b,G} = 1 \text{ if } b \in G \text{ and } b \notin (G_{1k} \cap G_{2k}).$$

[23] In the final dataset, zero-denominator cases are coded as 0 to preserve the full district universe.

earliest implementation milestone for which the district is exposed:

$$T_d = \begin{cases} t_{\text{F1}}, & \text{if } E_{d,\text{F1}}(\tau) > 0, \\ t_{\text{F2}}, & \text{if } E_{d,\text{F1}}(\tau) = 0 \;\&\; E_{d,\text{F2}}(\tau) > 0, \\ t_{\text{BN}}, & \text{if } E_{d,\text{F1}}(\tau) = E_{d,\text{F2}}(\tau) = 0 \;\&\; \text{BN}_d = 1, \\ \infty, & \text{if never treated,} \end{cases} \tag{4}$$

where $(t_{\text{F1}}, t_{\text{F2}}, t_{\text{BN}})$ are the start quarters for each phase (with Phase 1 becoming operational in 2023Q2, Phase 2 in 2023Q4, and the incorporation of Banco de la Nación (BN) into Phase 2 in January 2024). The absorbing treatment indicator is then:

$$D_{dt} = \mathbb{1}\{t \geq T_d\}. \tag{5}$$

Let Y_{idt} denote the outcome of individual i in district d and quarter t, with potential outcomes $Y_{idt}(1)$ and $Y_{idt}(0)$. The central object in Callaway and Sant'Anna (2021) is the cohort-by-time average treatment effect:

$$\text{ATT}(g,t) = \mathbb{E}[\, Y_{idt}(1) - Y_{idt}(0) \mid T_d = g \,], \qquad t \geq g. \tag{6}$$

From $\text{ATT}(g,t)$, I construct (i) a dynamic event-study profile by relative time $\ell = t - g$ and (ii) post-treatment averages (e.g., the average for $\ell \geq 0$), following the aggregation schemes proposed in Callaway and Sant'Anna (2021).

Under no anticipation and parallel trends for untreated outcomes, $\text{ATT}(g,t)$ is identified by comparing changes in outcomes for cohort g with units that are *not yet treated* at time t:

$$C_{g,t} = \{d : T_d > t\}. \tag{7}$$

Operationally, I estimate $\text{ATT}(g,t)$ using the Callaway and Sant'Anna (2021) estimator, which implements clean comparisons (treated units versus not-yet-treated units) and allows transparent aggregation in the presence of cohort and relative-time heterogeneity. Standard errors are clustered at the district level.

6 Results

This section summarizes the main results from the staggered difference-in-differences estimation, following Callaway and Sant'Anna (2021), of the Retail Payments Interoperability Strategy on (i) cash use, (ii) card usage, (iii) digital payments (internet banking and digital wallets) and financial inclusion, and (iv) a proxy for household financial fragility.[24]

6.1 Effects on cash use

Table 2 shows that interoperability reduces cash use in most expenditure categories. Because baseline cash use is high (pre-treatment means between 0.79 and 0.91 for frequent purchases), even declines of 1–5 pp represent economically meaningful shifts along the payment-choice margin.

The largest reductions are concentrated in recurring payments and services, where interoperability is most likely to relax salient frictions such as compatibility, coordination, and transfer convenience. Cash use falls by 4.4 pp in utilities (relative to a pre-treatment mean of 0.859) and by 5.2 pp in cooking fuels (relative to a mean of 0.794). In relative terms, these effects correspond to reductions of roughly 5–7% from pre-treatment levels, consistent with substitution away from cash in everyday transactions.

Significant declines are also observed in high-frequency consumption categories, including ready-to-eat food (-1.9 pp) and clothing and footwear (-3.1 pp), as well as smaller decreases in groceries, laundry, and personal hygiene (all between -1.5 and -1.7 pp). For Other, I do not detect statistically significant changes.

In practice, lowering compatibility frictions appears to expand the set of feasible digital transactions, raising the private return to paying digitally and accelerating substitution away from cash, particularly in categories dominated by repeated, low-value payments (Bech et al., 2018; Bianchi et al., 2023; Alvarez and Lippi, 2009).

6.2 Effects on card usage

Table 3 shows that, beyond cash, the use of cards (debit/credit) also declines in several categories. This pattern suggests that interoperability does not necessarily induce a simple "cash $\rightarrow$ card" substitution; instead, it appears to reallocate transactions toward other digital rails (e.g., instant

[24] Coefficients are interpreted as percentage-point (pp) changes in binary outcomes, with standard errors clustered at the district level. I report multiple-testing adjusted p-values using the Benjamini–Hochberg procedure (p *BH*), which I use as the benchmark for statistical significance.

Table 2: Interoperability and its effect on cash use by expenditure category

	Interoperability Exposure x Post	Std. Error	p-value	p BH	Pre-treatment mean
Groceries	-0.017*	0.009	0.057	0.071	0.911
Ready-to-eat food	-0.019**	0.008	0.023	0.045	0.909
Laundry	-0.017*	0.009	0.044	0.063	0.896
Utilities	-0.044***	0.011	0.000	0.001	0.859
Cooking fuels	-0.052***	0.013	0.000	0.000	0.794
Personal hygiene	-0.015*	0.007	0.044	0.063	0.910
Clothing and footwear	-0.031***	0.010	0.002	0.007	0.838
Furniture and appliances	-0.036*	0.021	0.082	0.091	0.253
Household appliances	-0.031**	0.013	0.017	0.042	0.106
Other	-0.006	0.014	0.663	0.663	0.836
N			233,196		
N Clusters			1,347		

Note: This table reports the effect of district-level exposure to interoperability on cash use by expenditure category. The exposure measure is constructed using district-level deposits prior to interoperability (Dec-2022) for institutions participating in each phase of the strategy. Outcomes are indicators equal to 1 if the respondent reports using cash for a given expenditure category. Estimation uses the staggered difference-in-differences approach of Callaway and Sant'Anna (2021). Standard errors are clustered at the district level. Statistical significance denoted by *, **, and *** is based on the adjusted p-value reported in column p BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the mean of each outcome in the pre-treatment period (before 2023Q2).

account-to-account transfers and interoperable wallets, often supported by QR codes or simple identifiers such as phone numbers) that compete directly with card payments in certain contexts.

Estimated effects are negative and statistically significant for groceries (-1.4 pp), laundry (-1.4 pp), utilities (-1.7 pp), cooking fuels (-1.0 pp), personal hygiene (-1.7 pp), and clothing and footwear (-1.3 pp). Although baseline card usage is low (between 2 and 6% depending on the category), the effects are large in relative terms. For instance, a 1.7 pp decline in utilities amounts to more than half of the pre-treatment mean (3.1%). For lower-frequency categories (furniture and appliances, household appliances), I do not detect meaningful changes.

A natural interpretation is that, by facilitating account-to-account and wallet-to-wallet payments via phone number or QR, interoperability reduces key frictions on the acceptance side and improves the user experience. These channels should be particularly relevant for small, repeated transactions. Merchants may tilt toward rails with lower acceptance costs or less reliance on spe-

cialized hardware, such as QR rather than POS terminals, while consumers value the convenience of a uniform payment experience regardless of provider (Allen et al., 2022; Alok et al., 2024). In the Peruvian case, BCRP documents emphasize that Phase 2 enabled QR interoperability and extended instant phone-based transfers to mobile banking apps, expanding use cases beyond wallets and intensifying competition across instruments (Fernández and García, 2024; García et al., 2025). Under this logic, simultaneous declines in both cash and cards in everyday spending categories are consistent with substitution toward a third instrument (interoperable transfers/payments), rather than a simple reshuffling between traditional means of payment.

Table 3: Interoperability and its effect on card usage by expenditure category

	Interoperability Exposure x Post	Std. Error	p-value	p BH	Pre-treatment mean
Groceries	-0.014**	0.006	0.019	0.039	0.055
Ready-to-eat food	-0.010	0.006	0.094	0.135	0.050
Laundry	-0.014**	0.006	0.018	0.039	0.047
Utilities	-0.017***	0.004	0.000	0.001	0.031
Cooking fuels	-0.010***	0.003	0.002	0.010	0.019
Personal hygiene	-0.017***	0.006	0.003	0.010	0.048
Clothing and footwear	-0.013**	0.006	0.027	0.044	0.054
Furniture and appliances	-0.003	0.003	0.416	0.520	0.012
Household appliances	0.001	0.003	0.685	0.761	0.012
Other	0.001	0.004	0.831	0.831	0.023
N			233,196		
N Clusters			1,347		

Note: This table reports the effect of district-level exposure to interoperability on the use of cards (debit and credit) by expenditure category. The exposure measure is constructed using district-level deposits prior to interoperability (Dec-2022) for institutions participating in each phase of the strategy. Outcomes are indicators equal to 1 if the respondent reports using at least one type of card for a given expenditure category. Estimation uses the staggered difference-in-differences approach of Callaway and Sant'Anna (2021). Standard errors are clustered at the district level. Statistical significance denoted by *, **, and *** is based on the adjusted p-value reported in column p BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the mean of each outcome in the pre-treatment period (before 2023Q2).

6.3 Financial inclusion, internet banking, and digital payments

Table 4 highlights a contrast between (i) the extensive margin of inclusion and (ii) the usage margin of digital channels. First, the effect on financial inclusion is negative but statistically indistinguish-

able from zero (-2.6 pp), which is plausible for two reasons. On the one hand, interoperability primarily relaxes usage frictions across existing networks, whereas opening and maintaining a formal account depends on additional barriers—documentation, perceived costs, trust, financial/digital literacy, and the supply of suitable products—that typically adjust more slowly than transactional margins (Allen et al., 2016; Ahamed and Mallick, 2019). On the other hand, within the phased rollout, expansion toward less banked segments may become more salient as additional participant types are incorporated (e.g., e-money) and onboarding processes improve. Indeed, institutional evidence suggests that the incorporation of actors such as Banco de la Nación or the maturation of Phase 3 (e-money/BIM) could be particularly relevant for expanding coverage, but its effects on broader outcomes may materialize with lags (Fernández and García, 2024; García et al., 2025).[25]

By contrast, I find an increase in internet banking usage of +3.3 pp (relative to a pre-treatment mean of 0.060).[26] Moreover, when I broaden the measure to capture combined use of internet banking and wallets (IB + Digital Wallets), the estimated effect rises to +5.0 pp In relative terms, these magnitudes imply increases on the order of 55–80% compared to pre-treatment levels, indicating that interoperability operates strongly on the intensive margin of payment digitalization.

This result is consistent with two complementary pieces of evidence. First, the international literature shows that shocks that reduce frictions—whether in adoption, acceptance, or coordination—can generate persistent increases in digital payment usage (Agarwal et al., 2024; Higgins, 2024). Second, descriptive evidence documents rapid and sustained growth in interoperable transactions following the rollout of Phases 1 and 2 as more entities and use cases were incorporated (Vásquez, 2024; García et al., 2025; Fernández and García, 2024).

6.4 Household financial fragility

Finally, Table 5 reports the effect on the proxy for household financial fragility. I find a marginally statistically significant reduction (at the 10% level) of 2.3–2.4 pp in the probability of fragility. Relative to a pre-treatment mean of 0.207, these effects correspond to a decline of roughly 11% in relative terms.

Although this evidence should be interpreted cautiously, the sign and stability across the two exposure measures are consistent with mechanisms documented for payment technologies and mobile money. By lowering transaction costs and facilitating timely transfers, households may

[25] As discussed in Section 3, by June 2025 Phases 1 and 2 accounted for 98.3% of all interoperable transactions, whereas Phase 3 accounted for the remaining 1.7% (García et al., 2025).

[26] As noted in Table 4, this estimate is restricted to 2023 due to changes in the INEI questionnaire in 2024.

Table 4: Interoperability and its effect on financial inclusion, internet banking usage, and digital payments

	Interoperability Exposure x Post	Std. Error	p-value	p BH	Pre-treatment mean	N	N Clusters
Financial Inclusion	-0.026	0.016	0.113	0.113	0.524	233,196	1,347
Internet Banking (IB)	0.033***	0.012	0.007	0.010	0.060	155,995	1,306
IB + Digital Wallets	0.050***	0.011	0.000	0.000	0.060	233,196	1,347

Note: This table reports the effect of district-level exposure to interoperability on financial inclusion, internet banking usage, and digital payments. For financial inclusion, the outcome is an indicator equal to 1 if the respondent reports having any type of bank account (financial inclusion). Internet banking usage is an indicator equal to 1 if the respondent reports using this channel for at least one of the nine expenditure categories. The measure IB + Digital Wallets additionally incorporates, for 2024, reported use of digital wallets. The exposure measure is constructed using district-level deposits prior to interoperability (Dec-2022) for institutions participating in each phase of the strategy. Estimation uses the staggered difference-in-differences approach of Callaway and Sant'Anna (2021). Starting in 2024, INEI expanded the internet-banking response option to explicitly include other channels (mobile banking and ATMs), so the estimation for internet banking is restricted to 2023 to preserve comparability. I also report estimates for IB + Digital Wallets, which incorporate INEI's new questions on mobile banking, ATMs, and digital wallets. Standard errors are clustered at the district level. Statistical significance denoted by *, **, and *** is based on the adjusted p-value reported in column p BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the mean of each outcome in the pre-treatment period (before 2023Q2).

smooth idiosyncratic shocks more quickly, reducing the need for precautionary dissaving or emergency borrowing (Jack and Suri, 2014; Riley, 2018). Interoperability may have strengthened informal insurance within social networks (family and friends) and improved short-run liquidity management. This interpretation is also consistent with the notion that cash remains dominant precisely in everyday, recurring payments. As these payments shift toward digital rails, households may face lower cash-management costs (time, transport, security risks) and greater capacity to receive and send funds immediately, both channels that plausibly translate into improved financial well-being (Lusardi et al., 2011; Bech et al., 2018).

7 Mechanisms

This section discusses plausible mechanisms that help interpret why interoperability may (i) reduce cash use, (ii) increase the use of digital channels and payments, and (iii) lower household financial fragility.

On the one hand, effective use of digital wallets and digital payments typically requires connectivity (a smartphone plus mobile data), so interoperability can raise the private return to maintaining mobile internet access. Table 6 shows that, in more exposed districts, the probability of reporting prepaid mobile internet increases by 7.5 to 7.9 pp (depending on the exposure measure), while fixed internet access declines by about 3.8 pp; I find no statistically precise effects on postpaid mobile internet. Because prepaid plans usually involve lower commitment and greater

Table 5: Interoperability and its effect on household financial fragility

	Financial Fragility
Interoperability Exposure 1 x Post	-0.023*
	(0.014)
Interoperability Exposure 2 x Post	-0.024*
	(0.014)
Pre-treatment mean	0.207
N	95,047
N Clusters	1,347

Note: This table reports the effect of district-level exposure to interoperability on a measure of household financial fragility. I estimate the effect using two exposure measures. The first measure (*Interoperability Exposure 1* $\times$ *Post*) is constructed using district-level deposits prior to interoperability (Dec-2022) for institutions participating in each phase of the strategy, while the second measure (*Interoperability Exposure 2* $\times$ *Post*) additionally incorporates the number of ATMs and banking agents at the same level. Financial fragility is an indicator equal to 1 if the respondent reports that, under the household's current economic situation, the household is forced to draw down savings or to borrow. Estimation uses the staggered difference-in-differences approach of Callaway and Sant'Anna (2021). Standard errors are clustered at the district level. Statistical significance denoted by *, **, and *** is based on the adjusted p-value reported in column p BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the mean of each outcome in the pre-treatment period (before 2023Q2).

flexibility for households facing tighter budget constraints, this pattern is consistent with an adjustment toward a "minimum digital capacity" needed to operate payments via mobile phones, particularly for P2P uses and low-value transactions.

Table 6: Effect of interoperability on household internet access by connection type

	Fixed Internet	Postpaid Mobile	Prepaid Mobile
Interoperability Exposure 1 x Post	-0.038*	0.024	0.079***
	(0.021)	(0.022)	(0.021)
Interoperability Exposure 2 x Post	-0.038*	0.008	0.075***
	(0.021)	(0.022)	(0.021)
Pre-treatment mean	0.376	0.551	0.548
N	142,366	141,255	141,255
N Clusters	1,303	1,303	1,303

Note: This table reports the effect of district-level exposure to interoperability on three measures of household internet access. I estimate effects using two exposure measures. The first measure (*Interoperability Exposure 1* $\times$ *Post*) is constructed using district-level deposits prior to interoperability (Dec-2022) for institutions participating in each phase of the strategy, while the second measure (*Interoperability Exposure 2* $\times$ *Post*) additionally incorporates the number of ATMs and banking agents at the same level. Outcomes are indicators equal to 1 if the household reports having fixed internet, postpaid mobile internet, or prepaid mobile internet access. Estimation uses the staggered difference-in-differences approach of Callaway and Sant'Anna (2021). Standard errors are clustered at the district level. Statistical significance denoted by *, **, and *** is based on the adjusted p-value reported in column p BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the mean of each outcome in the pre-treatment period (before 2023Q2).

This evidence should be interpreted cautiously—it does not establish a unique channel—but it is consistent with prior work documenting that shifts in the value of connectivity can translate into adoption of digital services and changes in economic behavior (Hjort and Poulsen, 2019; Aker, 2010). In our setting, interoperability increases the marginal utility of mobile data in districts with greater scope for cross-network matches, reinforcing the displacement from cash toward digital

payments documented in Tables 2 and 4.

As discussed earlier, retail payment markets often exhibit network externalities and coordination problems, meaning that the value of using a given instrument depends on how likely it is that the counterparty (a person or a merchant) can accept it and on how costly compatibility across networks is (Rochet and Tirole, 2002; Rysman, 2009). Interoperability targets this friction directly by increasing connectivity across ecosystems (initially P2P and, progressively, P2B), thereby raising the probability of "cross-network matches" within each district. In addition, implementation features that strengthen the coordination channel—with emphasis on end-user experience, monitoring of failed transactions, and operational follow-up—and efforts to expand the set of connected participants over time plausibly amplify the effects as the network grows (Vásquez, 2024; García et al., 2025; Aguirre, 2023).

A classic barrier to displacing cash is not only consumer preferences but also merchants' acceptance costs, including infrastructure requirements, fees, compatibility constraints, and learning costs (Allen et al., 2016, 2022). As interoperability extends to merchant payments and common standards consolidate, acceptance costs for small businesses fall because standardization replaces the need to display multiple provider-specific codes or solutions and enables a single acceptance point that works for users across different wallets and applications (García et al., 2024).

This mechanism is consistent with the fact that the largest cash effects concentrate in everyday and recurring payments (Table 2), where acceptance frictions and convenience tend to be first-order determinants of payment choice. It also helps interpret why, in some categories, declines in cash do not map into increased card usage (Table 3). If interoperable rails become relatively more convenient or cheaper to accept than traditional card infrastructure, they may displace both cash and cards in certain use cases (Bianchi et al., 2023; Alok et al., 2024).

The results on financial fragility (Table 5) are consistent with a liquidity-management and informal insurance channel. Greater payment connectivity may enable (i) more timely transfers within family and friend networks (P2P) and (ii) lower frictions in mobilizing liquid resources, for instance by receiving support during transitory shortfalls, which can reduce the need to draw down savings or borrow at the margin through improved risk-sharing and shock smoothing (Jack and Suri, 2014; Riley, 2018).

Beyond these mechanisms, both the literature and institutional evidence point to additional factors that may reinforce adoption and substitution away from cash, including (i) pricing structures and P2P fee waivers that facilitate learning and repeated use, (ii) improvements in user

experience and reductions in operational failures, (iii) trust, perceived security, and digital literacy, and (iv) the gradual entry of more participants and services that expand the set of use cases (Andia et al., 2025; Allen et al., 2016, 2022; Vásquez, 2024; García et al., 2025; Aurazo and Vega, 2021).

8 Robustness

This section assesses the sensitivity of the main findings along four dimensions: (i) the choice of estimator in staggered-adoption settings, (ii) an expanded exposure measure that combines deposits with physical financial infrastructure (ATMs and banking agents), (iii) the choice of the pre-treatment date used to construct exposure, and (iv) plausible sources of time-coincident confounding. I also report the results of a placebo test.

(i) Estimator choice. Given the well-known limitations of TWFE under staggered adoption with heterogeneous effects, I replicate the baseline estimates using the approaches proposed by Sun and Abraham (2021), Borusyak et al. (2024), and de Chaisemartin and D'Haultfœuille (2024). The goal is to verify that the results are not driven by a particular implicit weighting scheme or by restrictive homogeneity assumptions. The main conclusions—a decline in cash use, an increase in the use of digital channels/payments, and a marginal reduction in financial fragility—are robust across estimators (see Figures A.6 through A.11). In particular, the event-study profiles show (i) no systematic pre-trends in the leads, consistent with parallel trends, and (ii) a gradual build-up of effects in the post-treatment period.

(ii) Expanded exposure measure based on deposits and physical financial infrastructure. Because effective adoption of digital payments may be shaped by cash-in/cash-out frictions and access to financial services, especially in districts with sparse infrastructure, I replace the baseline exposure measure (pre-treatment deposits) with an alternative measure that incorporates, in addition to deposits, the number of ATMs and banking agents associated with the institutions in each block.[27] This expanded measure is designed to capture not only the stock of deposits that could plausibly be mobilized through interoperable payments, but also the operational capillarity that facilitates funding, withdrawals, and day-to-day use of financial services.

[27] Panel (b) of Figure A.4 displays the spatial distribution of this alternative exposure measure. Panel (b) of Figure A.5 shows the evolution of the evaluation panel, tracking treated individuals by phase.

Starting from the deposits-based exposure measure $E_{dk}(\tau)$ in equation (3), I construct an analogue based on physical infrastructure. For each phase $k \in \{\text{F1}, \text{F2}, \text{BN}\}$ and each block $G \in \{G_{1k}, G_{2k}\}$, I define the "mass" of access points associated with block G in district d as

$$C_{dk}^{G}(\tau) = \sum_{b \in G} w_{b,G}\, \text{AP}_{db\tau}, \tag{8}$$

where $\text{AP}_{db\tau}$ denotes the number of ATMs and banking agents of institution b in district d, measured at $\tau =$ December 2022, and $w_{b,G}$ are the same weights used in (1) to handle institutions that belong to both blocks. The relative share of block G_{1k} in terms of physical infrastructure is defined as

$$\tilde{s}_{dk}(\tau) = \frac{C_{dk}^{G_{1k}}(\tau)}{C_{dk}^{G_{1k}}(\tau) + C_{dk}^{G_{2k}}(\tau)}. \tag{9}$$

Analogously to (3), I define an infrastructure-based exposure index as

$$\tilde{E}_{dk}(\tau) = 4\,\tilde{s}_{dk}(\tau)\left[1 - \tilde{s}_{dk}(\tau)\right] \in [0, 1], \tag{10}$$

which is maximized when both blocks have a balanced presence of access points in the district and converges to zero when one block dominates or when infrastructure associated with both blocks is essentially absent (in that case, I set $\tilde{E}_{dk}(\tau) = 0$).

To combine deposits and physical infrastructure in a parsimonious way, I standardize both phase-specific exposure indices and construct, at the district level, a composite measure that averages the two dimensions. Let $\bar{E}_k$ and $\sigma(E_k)$ denote the mean and standard deviation of $E_{dk}(\tau)$ across districts in phase k, and let $\bar{\tilde{E}}_k$ and $\sigma(\tilde{E}_k)$ denote the analogous objects for $\tilde{E}_{dk}(\tau)$. I then define the expanded exposure measure as

$$E_{dk}^{\text{expanded}}(\tau) = \frac{1}{2}\left(\frac{E_{dk}(\tau) - \bar{E}_k}{\sigma(E_k)} + \frac{\tilde{E}_{dk}(\tau) - \bar{\tilde{E}}_k}{\sigma(\tilde{E}_k)}\right). \tag{11}$$

By construction, $E_{dk}^{\text{expanded}}(\tau)$ takes high values in districts where (i) deposits are relatively balanced across the two blocks that begin to interoperate and (ii) the ATM/correspondent infrastructure associated with those blocks is relatively dense. Conversely, the index approaches zero or becomes negative in districts where exposure through deposits and/or infrastructure is low or highly skewed toward one block. I use $E_{dk}^{\text{expanded}}(\tau)$ as an alternative pre-treatment exposure

measure to define district treatment cohorts in the staggered difference-in-differences design.[28]

In terms of interpretation, this expanded exposure measure differs from the baseline in how the set of treated districts is defined. Under the main specification, a district is considered exposed in phase k whenever the deposits-based exposure $E_{dk}(\tau)$ is strictly greater than zero, i.e., whenever there is any economically meaningful mass associated with the blocks that begin to interoperate. Under the expanded measure, I use $E_{dk}^{\text{expanded}}(\tau)$ and classify a district as treated when this composite index is greater than zero, which effectively selects districts with combined exposure (deposits and infrastructure) above the national mean for the corresponding phase. Thus, while the baseline measure separates districts along an extensive margin of exposure, the expanded measure emphasizes districts with high relative exposure, allowing me to assess robustness by focusing on locations where the presence of participating institutions is strongest in both financial and operational terms.

At an aggregate level, the pattern of results is unchanged: (i) cash use declines for everyday transactions—particularly utilities and cooking fuels (Table A.2); (ii) card use falls in categories where interoperable wallets/transfers plausibly compete with traditional acquiring infrastructure (Table A.3); (iii) the use of digital channels rises—internet banking and wallets (Table A.4); (iv) financial fragility declines marginally (Table 5); and (v) prepaid mobile internet access increases (Table 6). The consistency in signs and orders of magnitude across exposure measures indicates that the findings do not hinge on a single proxy for exposure and alleviates concerns that the estimates are merely capturing pre-existing differences in physical capillarity.

(iii) Choice of the pre-treatment date used to construct exposure. As an additional robustness check, I reconstruct the exposure measure using deposits as of March 2023, the last month prior to the start of interoperability in Phase 1. The purpose of this exercise is to show that the exposure variable is stable over a reasonable pre-treatment window. In practice, results based on March 2023 deposits closely mirror the baseline results using December 2022 deposits (see Tables A.5, A.6, and A.7).

(iv) *Not-yet-treated* controls versus never-treated controls. As a further robustness exercise, I restrict the sample to districts with at least one ATM or banking agent to improve comparability

[28] The composite measure preserves the differential exposure structure of the baseline design while explicitly incorporating local access to physical financial infrastructure, which is plausibly first-order for the adoption and effective use of digital payments.

between treated and comparison units. The qualitative patterns remain unchanged, suggesting that the changing composition of the control group is not driving the findings. For details, see Tables A.8, A.9, and A.10.

(v) Multiple-testing correction. Because I estimate multiple outcomes by spending category, all tables report *p*-values adjusted using Benjamini and Hochberg (1995), and I primarily assess statistical significance based on that column. This criterion is especially relevant in Tables 2–4, where closely related margins (spending categories) are tested and the goal is to mitigate false positives arising from multiplicity.

(vi) Placebo test. As an additional validation of the identification strategy, I conduct a placebo exercise in which I restrict the sample to the pre-treatmen period, from 2021Q3 through 2023Q1, and assign placebo treatment dates to the three district cohorts. Specifically, I impute placebo adoption dates of 2022Q1, 2022Q2, and 2022Q3 for the cohorts that in the actual implementation begin operating in 2023Q2, 2023Q4, and 2024Q1, respectively. I keep the same predetermined exposure measure and estimate the regressions using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021). For cash use (Table A.11), placebo coefficients are small and statistically indistinguishable from zero in almost all categories. Only the "Other" category exhibits a positive and significant coefficient, which is modest in magnitude and opposite in sign relative to the negative effects in the baseline specifications, suggesting an isolated finding likely attributable to sampling noise. For card use (Table A.12), none of the placebo coefficients are significant either before or after multiple-testing adjustment (Benjamini and Hochberg, 1995), and point estimates are very close to zero across all spending categories. Similarly, Table A.13 shows that placebo estimates for financial inclusion, internet banking, and digital payments are very small and not statistically significant, even without multiplicity corrections. Taken together, the absence of systematic placebo effects indicates that district-level exposure to interoperability is not proxying for pre-existing trends or anticipated shocks over 2021Q3–2023Q1, thereby reinforcing the causal interpretation of the estimated effects in the main tables.

9 Heterogeneity

The effects of interoperability need not be spatially homogeneous. In payments markets characterized by network externalities and coordination frictions, adoption and effective use of digital in-

struments depend on complementarities with local conditions such as connectivity infrastructure, human capital, trust, and the degree of formalization of economic activity. In Peru, sizable gaps in access to financial services and in the use of digital channels persist, reflecting perceived costs, liquidity constraints, and geographic barriers (Demirgüç-Kunt et al., 2022; García and Andía, 2022; Vega and Vásquez, 2022). This section therefore examines whether the interoperability strategy induces stronger substitution away from cash and greater adoption of digital channels in districts that were better positioned ex ante in terms of connectivity, human capital, and formality, and whether effects are muted in districts with high exposure to social programs, where digital-skills constraints and adoption costs tend to be more salient.

To proxy for districts' initial conditions, I construct a set of district-level variables measured in 2022 (pre-treatment). These covariates capture four dimensions: (i) human capital, via the educational composition (shares with primary, secondary, and tertiary education); (ii) digital infrastructure and usage, via internet access and use, cellphone access, and ownership of a personal cellphone; (iii) socioeconomic characteristics, including poverty, informality, urbanization, and the age–sex composition; and (iv) exposure to social programs, proxied by the share of the population reporting income from Juntos or Pensión 65.

For each covariate Z_d I define a dummy $dZ_d = \mathbb{1}\{Z_d \geq \text{median}(Z)\}$, where the median is computed across districts with available 2022 information. This approach has two practical advantages. First, it relies exclusively on pre-treatment information, which limits concerns that heterogeneity patterns are contaminated by endogenous changes induced by interoperability itself. Second, it yields simple and comparable partitions across a large set of dimensions. A drawback is that median splits discard intensity information; accordingly, I interpret these results as descriptive evidence on complementarities and frictions rather than as structural thresholds.

To estimate heterogeneous effects, I use the difference-in-differences estimator of Borusyak et al. (2024), which imputes counterfactual outcomes for treated units based on the trajectory of not-yet-treated districts and computes the post-treatment effect as the average gap between observed outcomes and the imputed counterfactual within each group (high vs. low) defined by pre-treatment characteristics. Coefficients are interpreted as percentage-point changes in the probability of the outcome.

Motivated by the main findings, I focus on four outcomes: (i) cash use for utilities, (ii) cash use for household fuels, (iii) internet-banking use, and (iv) digital-payments use.

Cash substitution in household expenditures. Figures A.13 and A.14 document pronounced heterogeneity in the reduction of cash use for utilities and household fuels. The decline in cash is concentrated in districts with stronger pre-treatment connectivity and digital capabilities, while it is substantially smaller in more informal districts and in districts with greater exposure to social programs. Taken together, this pattern suggests that interoperability relaxes technological frictions, but its real effects hinge on constraints to access and usage, as well as on incentives linked to payment traceability.

For housing (Figure A.13) and fuels (Figure A.14), the reduction in cash use is concentrated in highly connected districts. In those areas, estimates range between -5.4 and -5.8 pp, whereas in low-connectivity districts effects are close to zero and statistically indistinguishable from it. Heterogeneity by human capital is consistent with this pattern, as effects are larger in districts where the share of the population with only primary education is lower. Overall, these results indicate that interoperability reduces cash use primarily when digital-access constraints and human-capital limitations are less binding (Andia et al., 2025).

Heterogeneity by informality is stark and points to a traceability mechanism. For housing and fuels, districts with high informality exhibit small or statistically insignificant effects (approximately -0.7 to -1.4 pp), while districts with low informality display large reductions in cash use (approximately -5.3 to -5.8 pp). This pattern is consistent with the idea that, in more informal environments, the private value of cash—as an instrument that preserves anonymity and reduces traceability—limits substitution toward recordable payment methods even when interoperability lowers technological frictions (Rogoff, 2016; Immordino and Russo, 2018; Kleven et al., 2011; Pomeranz, 2015; Naritomi, 2019).

Urbanization amplifies effects, but in a category-specific way. For utilities, the reduction in cash is concentrated in more urban districts and becomes negligible in less urban ones, whereas for fuels the decline is sizable in both groups, suggesting a more homogeneous acceptance structure in that market. Greater exposure to social programs is also associated with weaker cash substitution. Districts with high coverage of Juntos or Pensión 65 exhibit small or statistically insignificant declines, in contrast to reductions of roughly -5.4 to -5.8 pp in districts with low coverage, a pattern consistent with adoption frictions and a stronger preference for immediate liquidity (cash-out) (Muralidharan et al., 2016).

Adoption of internet banking and digital payments. Figures A.15 and A.16 show that interoperability increases the use of digital channels, again with strong heterogeneity by digital infrastructure, human capital, formality, and urbanization.

For internet banking (Figure A.15), impacts are concentrated in districts with stronger pretreatment digital infrastructure and capabilities. In particular, in districts with high internet access or usage, and with higher ownership of a personal cellphone, internet-banking use rises by about +7.4 to +7.7 pp, whereas in low-connectivity districts effects are small and statistically indistinguishable from zero. For digital payments (Figure A.16), the increase is more widespread but preserves similar gradients. Highly connected districts experience increases around +9.4–+9.6 pp, relative to much smaller—and often insignificant—effects in low-connectivity districts. Effects are also larger where relative human capital is higher, proxied by a lower share with only primary schooling, suggesting that learning and adoption costs remain relevant even as interoperability reduces transactional frictions.

Informality dampens growth along both digital margins. In high-informality districts, increases in digital payments and internet banking are modest—around +2.1 pp and +1.0 pp, respectively—in contrast to substantial increases in low-informality districts, on the order of +9.4 pp and +7.5 pp Finally, urbanization emerges as a central determinant of adoption. Effects are large and positive in more urban districts but turn negative in less urban districts, a pattern consistent with the view that outside urban areas, constraints in local acceptance, connectivity, and learning may prevent interoperability from translating into effective use, or may redirect transactions toward instruments not fully captured by the outcomes analyzed.

10 Conclusions

This paper evaluates the effects of the BCRP's Retail Payments Interoperability Strategy on payment-instrument choice and on welfare-relevant margins of household financial well-being in Peru. The results point to meaningful substitution away from cash in everyday, recurrent expenditures, with economically sizable declines in utilities and cooking fuels. I also document a reduction in card use, suggesting that interoperability does not merely reallocate transactions across traditional instruments, but rather shifts activity toward interoperable digital rails that compete with conventional acquiring infrastructure in small, repeated transactions.

In parallel, interoperability increases the use of digital channels, most notably internet banking

and an aggregate measure of digital payments, while I find no robust effects on the extensive margin of financial inclusion. This pattern is consistent with a policy that lowers compatibility and coordination frictions in the use of existing instruments without necessarily removing more structural barriers to opening and maintaining formal accounts. Along welfare dimensions, the paper provides suggestive evidence of short-run improvements, reflected in a marginal decline in household financial fragility, in line with liquidity-management and risk-sharing mechanisms operating through more timely transfers and lower transaction costs.

The evidence on mechanisms and heterogeneity further supports this interpretation. In more exposed districts, mobile prepaid internet access increases while fixed connections decline, with no clear changes in postpaid access, a pattern consistent with households acquiring a minimum level of connectivity needed to transact via mobile devices. Effects are larger in districts with stronger pre-treatment connectivity, higher human capital, and greater formality, whereas impacts tend to be smaller or statistically indistinguishable from zero in more informal, less urban areas, or in districts with greater exposure to social programs. Overall, the evidence points to complementarities, given that interoperability reduces compatibility and coordination frictions, but its translation into effective use depends on local constraints of connectivity, learning, and acceptance.

Two limitations bound the scope of the findings. First, ENAHO records digital-wallet use only from 2024 onward, so there is no pre-treatment baseline that would allow me to assess parallel trends and estimate the causal effect on that outcome directly. Second, the lack of granular territorial information on electronic-money issuers limits the analysis of Phase 3, although this constraint is less consequential given the relatively limited scale of Phase 3 through 2025, as documented in earlier sections.

From a policy perspective, interoperability modernized the payments system by reducing coordination frictions and expanding digital transactions, potentially accelerating the transition away from cash. At the same time, the evidence indicates that impacts were heterogeneous and constrained by gaps in connectivity and digital capabilities, high informality, and adoption and learning costs. Interoperability therefore appears as a necessary condition whose effects are amplified by complementary investments in digital infrastructure, financial and digital literacy, and policies that reduce incentives to remain informal.

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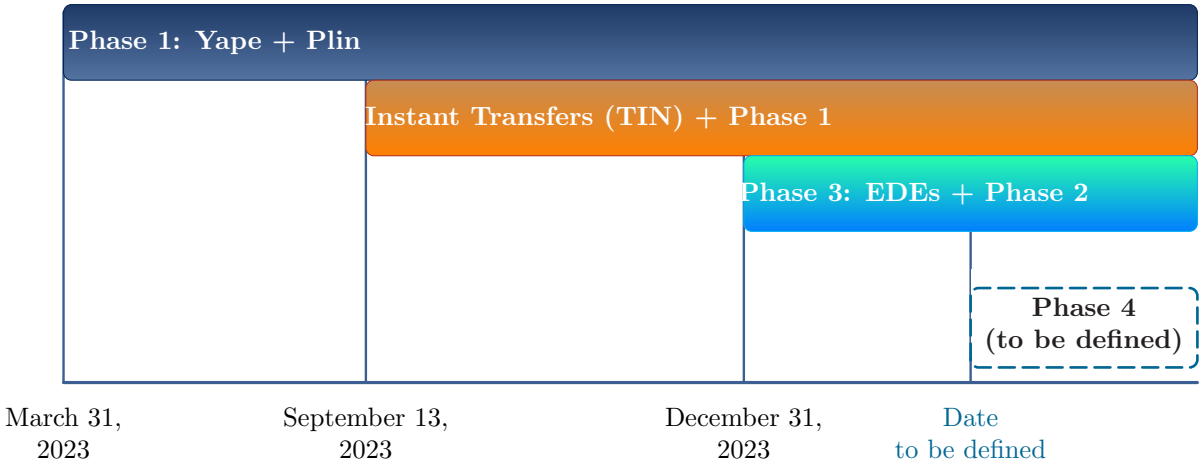
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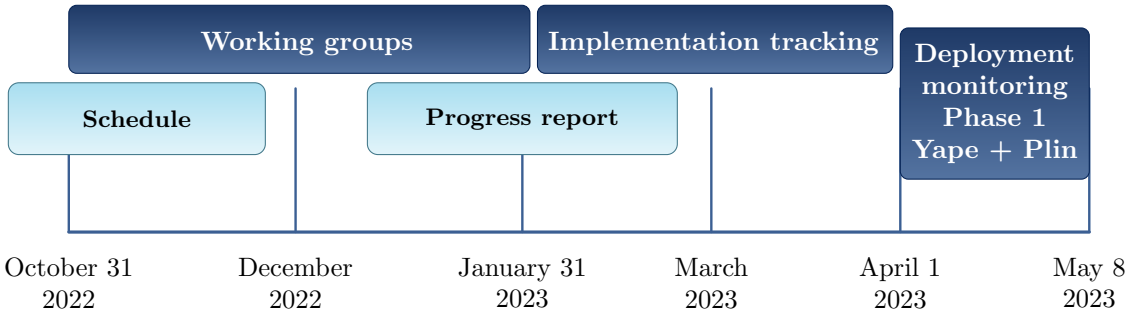
A Additional figures and tables

Figure A.1: Timeline of the implementation of the Retail Payments Interoperability Strategy



Source: Author's elaboration based on BCRP reports.

Figure A.2: Phase 1 implementation



Source: Author's elaboration based on BCRP reports.

Exposure measure: To illustrate the construction, consider the district of Chachapoyas (province of Chachapoyas, Amazonas region), which had 12 financial institutions (banks, municipal savings banks, and finance companies). Of these, 4 institutions were *Yape* participants, 2 were *Plin* participants, one institution participated in both networks, and 5 institutions participated in neither. The first step in the exposure measure is the weighted sum of district-level deposits for each group: in this district, I construct two deposit masses, one for the *Yape* block and one for the *Plin* block.[29]

Using these masses, I compute each group's relative share and then the exposure index defined in Equations 2 and 3, which takes values in $[0, 1]$. High values indicate a more balanced

[29] For institutions that participate in both *Yape* and *Plin*, I use a weight of 0.5 in each group so that the exposure index captures the possibility of interoperability for users of these institutions without double-counting their deposits.

distribution of deposits across groups and therefore greater interoperability potential; low values reflect concentration in a single group and lower (though not necessarily zero) potential. Exposure equals zero when one group concentrates 100% of deposits and also when there is no deposit information. To mitigate the latter limitation, as a robustness check I construct an alternative measure that incorporates, in addition to deposits, the number of ATMs and banking agents, capturing exposure in districts without deposits reported but with relevant financial infrastructure.

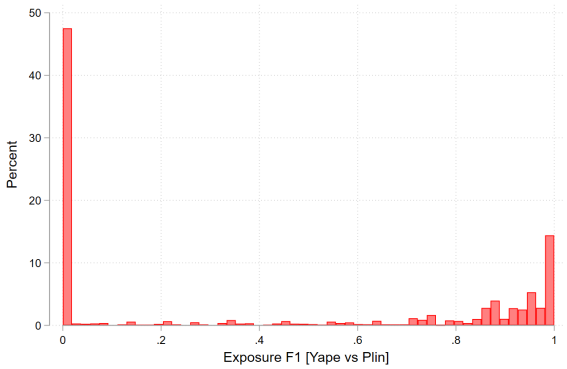
In Phase 2, district-level exposure is also constructed from two groups. The first group concentrates the deposit mass of Phase 1 participant institutions in the predetermined period, while the second group aggregates the deposit mass of institutions that enter as Phase 2 participants. This definition captures that districts not exposed in Phase 1 can become exposed in Phase 2 as new institutions join the interoperability strategy.

For the Banco de la Nación entry phase, I follow an analogous procedure. Group 1 concentrates the deposit mass of institutions participating in Phases 1 and 2, and Group 2 includes exclusively the deposit mass of Banco de la Nación, capturing exposure associated with its incorporation in January 2024. The resulting distribution (Panel (c) of Figure A.3) shows that a large fraction of districts have zero exposure, mainly because they do not register Banco de la Nación deposits or, in some cases, only register deposits from that institution, implying no possibility of cross-block interoperability. This pattern is consistent with Banco de la Nación's broader territorial coverage in districts where other institutions do not operate; in contrast, in Phases 1 and 2 the presence of more participant institutions in both groups tends to increase district-level interoperability potential.

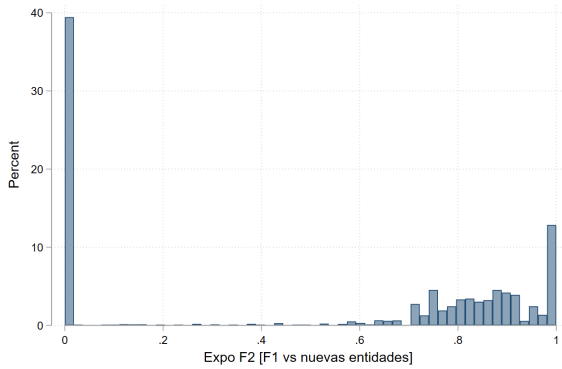
Exposure measures are constructed independently by phase and then translated into a staggered treatment variable following Equations 4 and 5. Specifically, a district is classified as treated in Phase 1 when its exposure measure is positive. For Phase 2, a district is considered treated if it has positive exposure and was not treated in Phase 1. Similarly, in the Banco de la Nación phase, a district is treated when its exposure is positive and it was not treated in earlier phases. The resulting staggered assignment is illustrated in Figure A.5.

Figure A.3: Distribution of Exposure Measure

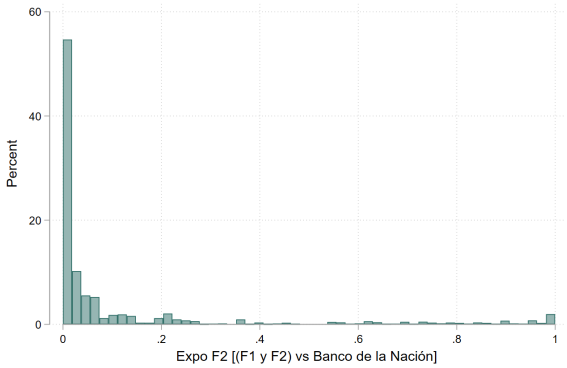
(a) Phase 1



(b) Phase 1 + Phase 2



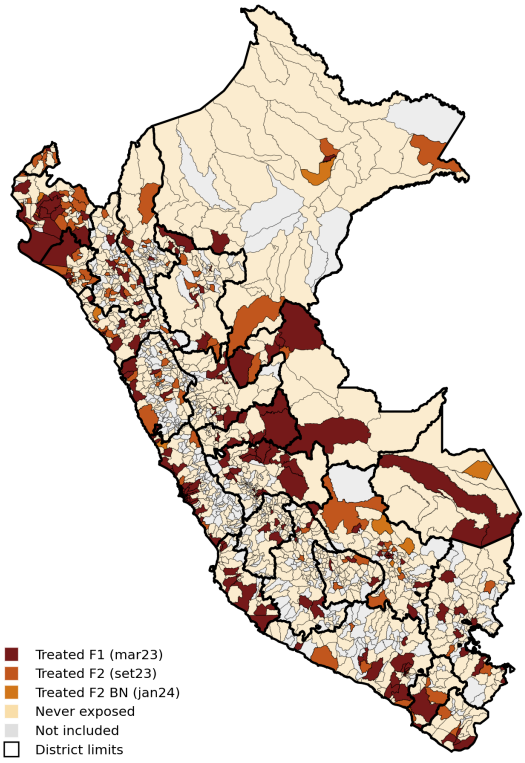
(c) Phase 1 + Phase 2 + Banco de la Nación



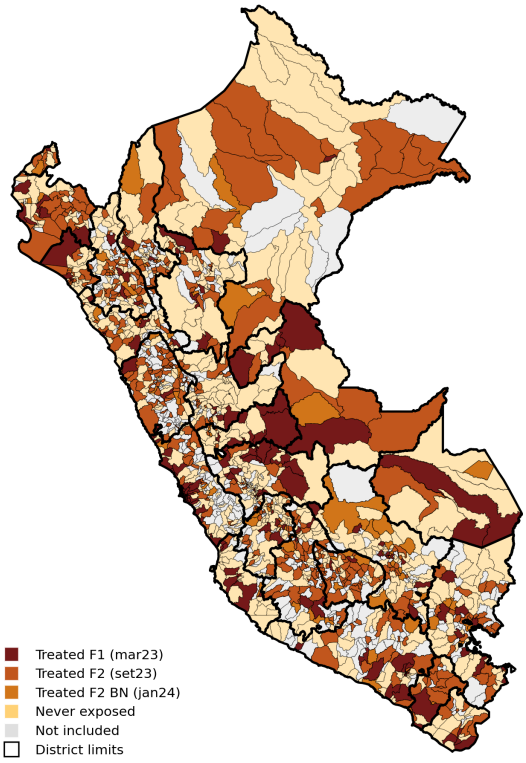
Note: This figure shows the distribution of district-level exposure, computed using district deposits for the groups of institutions participating in each interoperability phase. Panel (a) displays the distribution of the Phase 1 exposure measure, using deposits of institutions participating in this phase (Yape and Plin users). Panel (b) displays the distribution of the Phase 2 exposure measure, using the deposit mass of financial institutions participating in Phase 1 and the institutions that enter in Phase 2, in order to capture interoperability at this stage. Finally, Panel (c) displays the distribution of the exposure measure incorporating Banco de la Nación, using the deposit mass of institutions participating in the previous phases and Banco de la Nación.

Figure A.4: Map: district-level exposure

(a) Exposure measurement 1



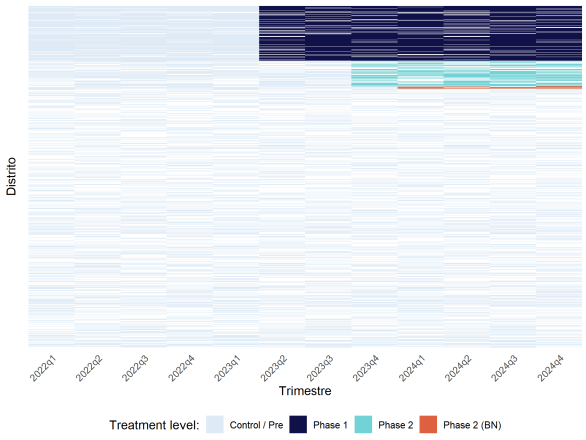
(b) Exposure measurement 2



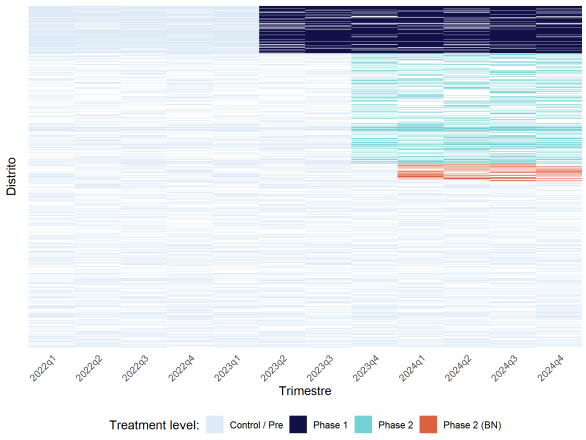
Note: The figure shows the distribution of exposure at the district level, in its two versions. Exposure measure 1 (Panel a) is calculated using district-level deposits of the groups of institutions participating in each phase of interoperability, while the second exposure measure (Panel b) includes, in addition to deposits, the number of ATMs and banking agents.

Figure A.5: Quarterly district panel assessment (2022q1-2024q4)

(a) Exposure measurement 1



(b) Exposure measurement 2



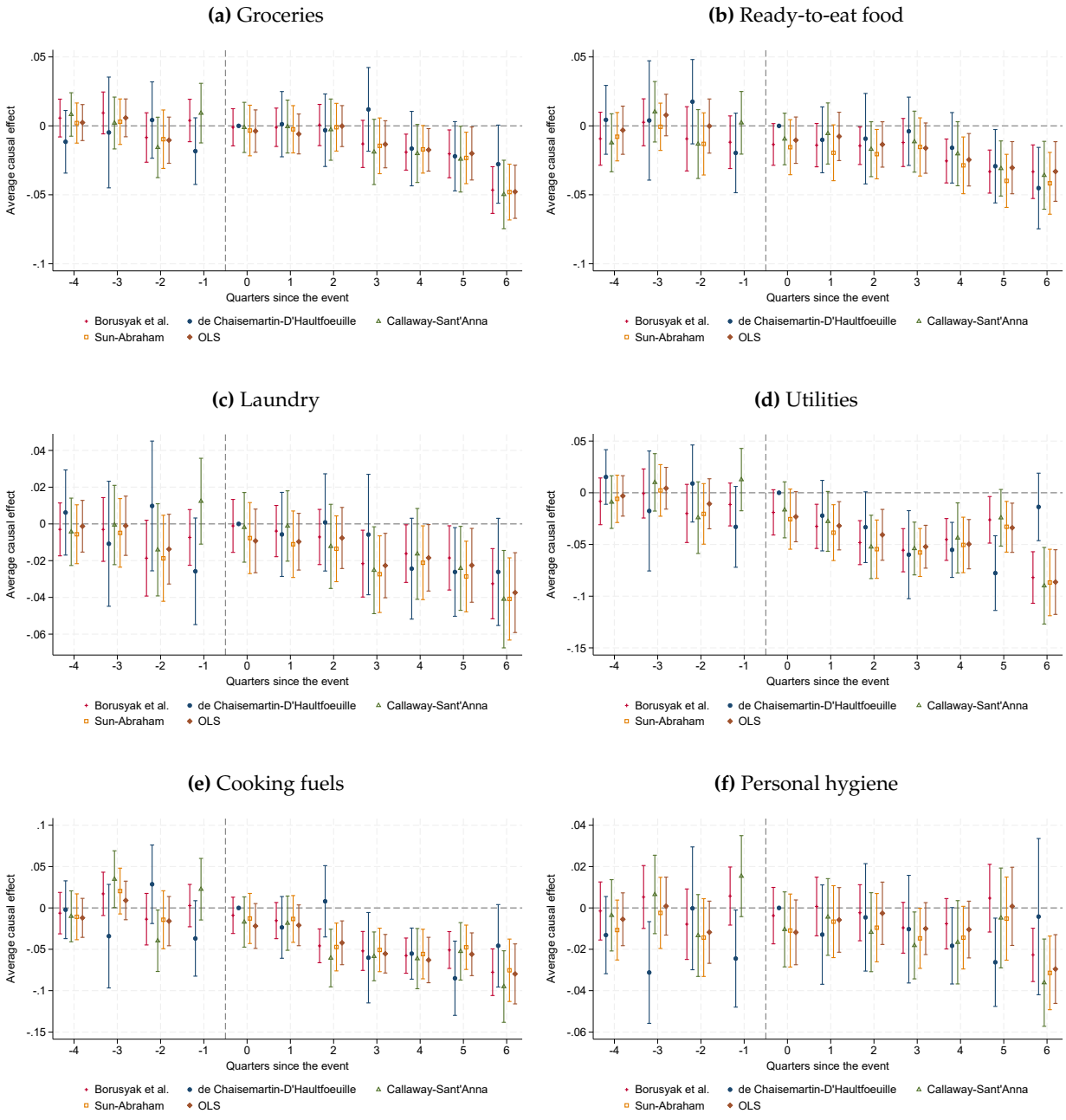
Note: Own work based on the PanelView package in R, developed by Mou et al. (2023).

Table A.1: Entities participating in the retail payments interoperability strategy

Phase 1: Yape-Plin Interoperability				
Type of institution	Entity	Yape user	Plin user	Start Phase 1
Bank	Banco de Crédito del Perú (BCP)	X		31 mar. 2023
Bank	Mibanco	X		31 mar. 2023
Bank	BBVA		X	31 mar. 2023
Bank	Interbank		X	31 mar. 2023
Bank	Scotiabank Perú		X	31 mar. 2023
Bank	Alfin Banco		X	31 mar. 2023
Bank	BanBif		X	31 mar. 2023
CMAC	CMAC Arequipa		X	31 mar. 2023
CMAC	CMAC Huancayo	X	X	31 mar. 2023
CMAC	CMAC Ica	X	X	31 mar. 2023
CMAC	CMAC Piura	X		31 mar. 2023
CMAC	CMAC Tacna	X		31 mar. 2023
CMAC	CMAC Trujillo	X		31 mar. 2023
CMAC	CMCP Lima (Caja Metropolitana)	X		31 mar. 2023
Financiera	Financiera Confianza		X	31 mar. 2023
Phase 2: Instant transfers and interoperable QR codes				
Type of institution	Entity	Phase 2 Participant		Start of Phase 2
Bank	Banco de Comercio	X		13 sep. 2023
Bank	Banco Pichincha	X		13 sep. 2023
Bank	Banco GNB	X		13 sep. 2023
Bank	Banco Falabella	X		13 sep. 2023
Bank	BanBif	X		13 sep. 2023
Bank	Alfin Banco	X		13 sep. 2023
Bank	BBVA	X		13 sep. 2023
Bank	Banco de Crédito del Perú (BCP)	X		13 sep. 2023
Bank	Interbank	X		13 sep. 2023
Bank	Mibanco	X		13 sep. 2023
Bank	Scotiabank Perú	X		13 sep. 2023
Bank	Citibank del Perú	X		13 sep. 2023
Bank	Banco de la Nación	X		Ene 2024
CMAC	CMAC Arequipa	X		13 sep. 2023
CMAC	CMAC Cusco	X		13 sep. 2023
CMAC	CMAC Ica	X		13 sep. 2023
CMAC	CMAC Piura	X		13 sep. 2023
CMAC	CMAC Sullana	X		13 sep. 2023
CMAC	CMAC Trujillo	X		13 sep. 2023
Financiera	Crediscotia	X		13 sep. 2023
Financiera	Financiera Oh!	X		13 sep. 2023
Financiera	Compartamos Banco	X		13 sep. 2023

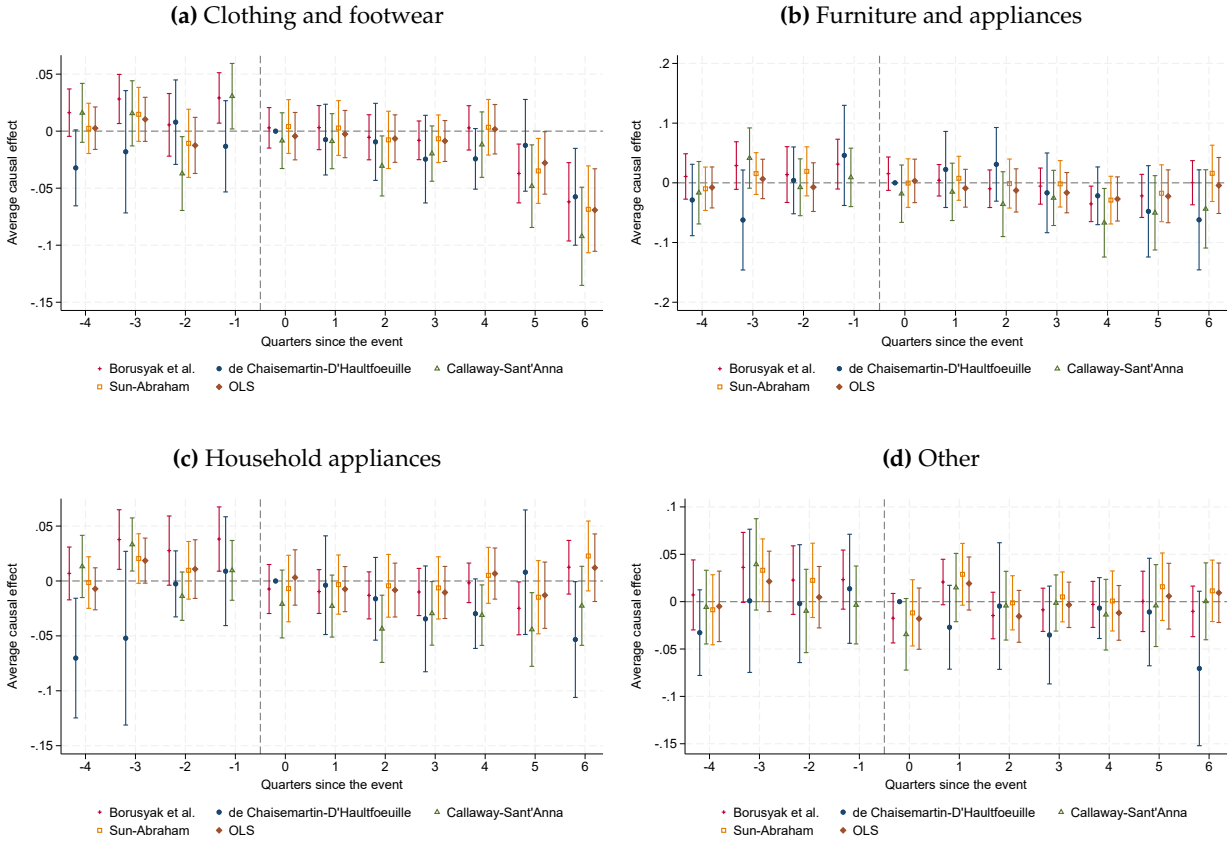
Note: Own elaboration based on the Retail Payments Interoperability Strategy (Circular N.º 024-2022-BCRP) and progress reports from the García et al. (2024); García et al. (2025).

Figure A.6: Dynamic effects on cash use by expenditure category



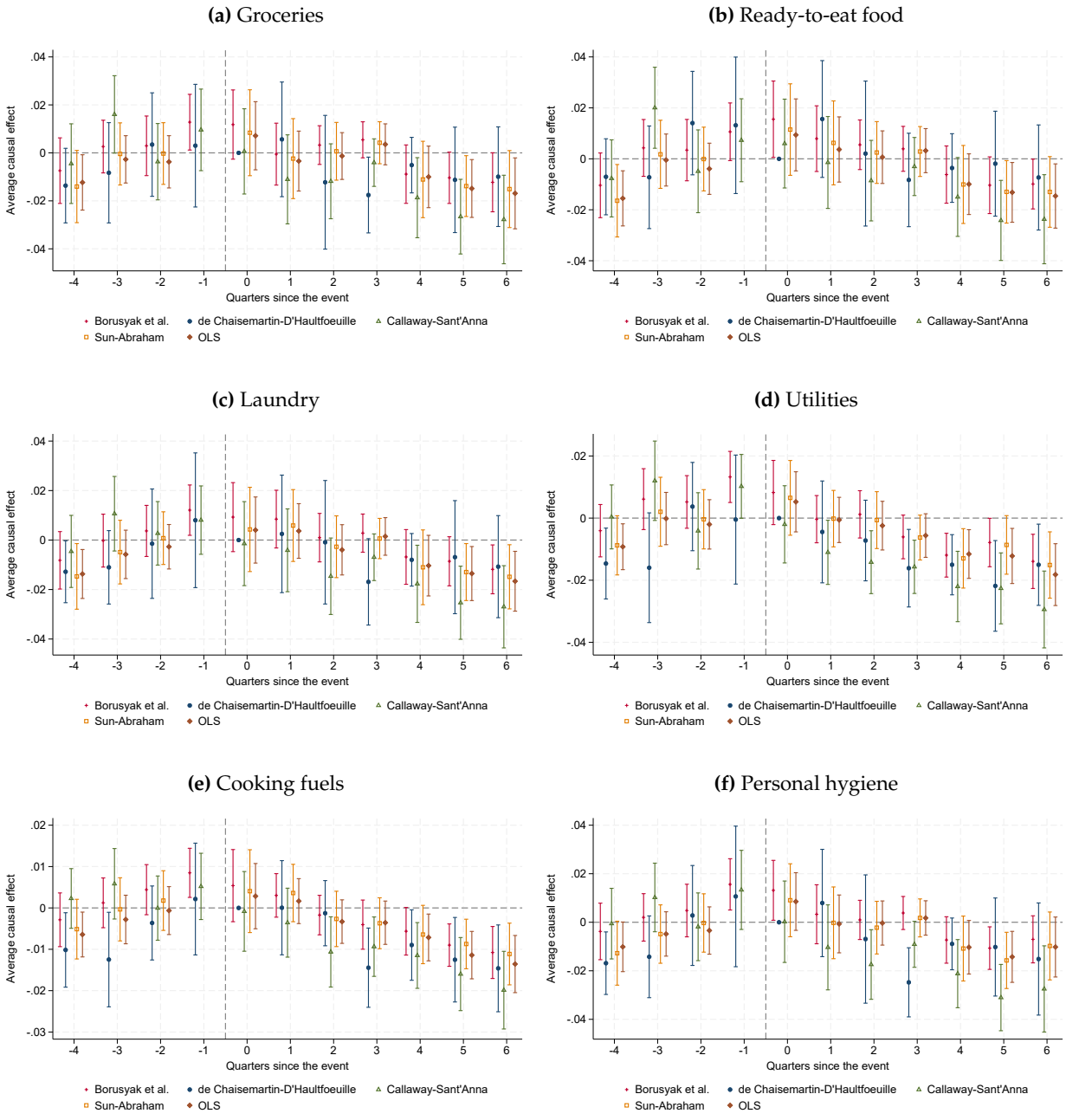
Note: This figure reports dynamic effects of district-level exposure to interoperability on cash use by expenditure category. The exposure measure is constructed from district-level deposits in the pre-interoperability period (December 2022) for institutions participating in each phase of the strategy. Outcomes are indicators equal to one if the individual reports using cash for the corresponding expenditure category. Estimation uses staggered-adoption difference-in-differences methods following Callaway and Sant'Anna (2021); de Chaisemartin and D'Haultfœuille (2024); Borusyak et al. (2024); Sun and Abraham (2021). Standard errors are clustered at the district level.

Figure A.7: Dynamic effects on cash use by expenditure category



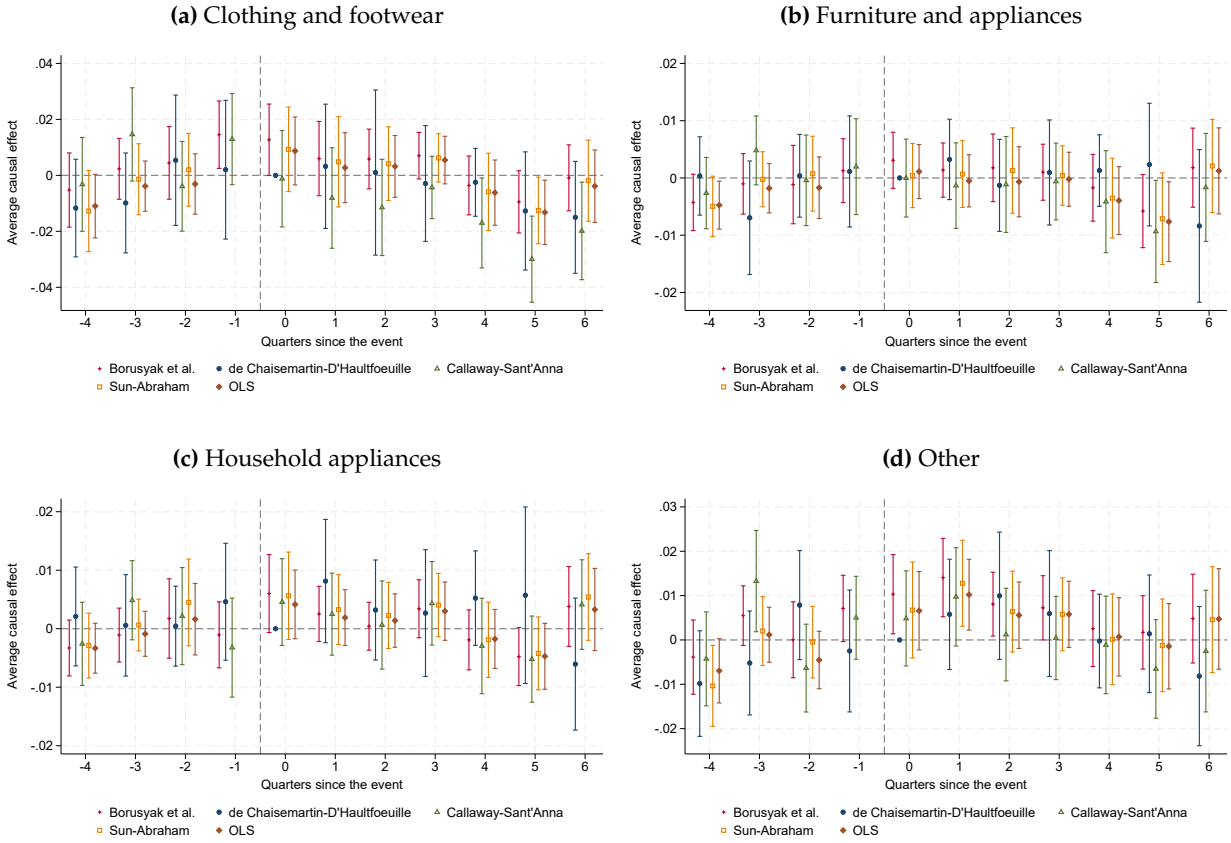
Note: This figure reports dynamic effects of district-level exposure to interoperability on cash use by expenditure category. The exposure measure is constructed from district-level deposits in the pre-interoperability period (December 2022) for institutions participating in each phase of the strategy. Outcomes are indicators equal to one if the individual reports using cash for the corresponding expenditure category. Estimation uses staggered-adoption difference-in-differences methods following Callaway and Sant'Anna (2021); de Chaisemartin and D'Haultfœuille (2024); Borusyak et al. (2024); Sun and Abraham (2021). Standard errors are clustered at the district level.

Figure A.8: Dynamic effects on card use by expenditure category



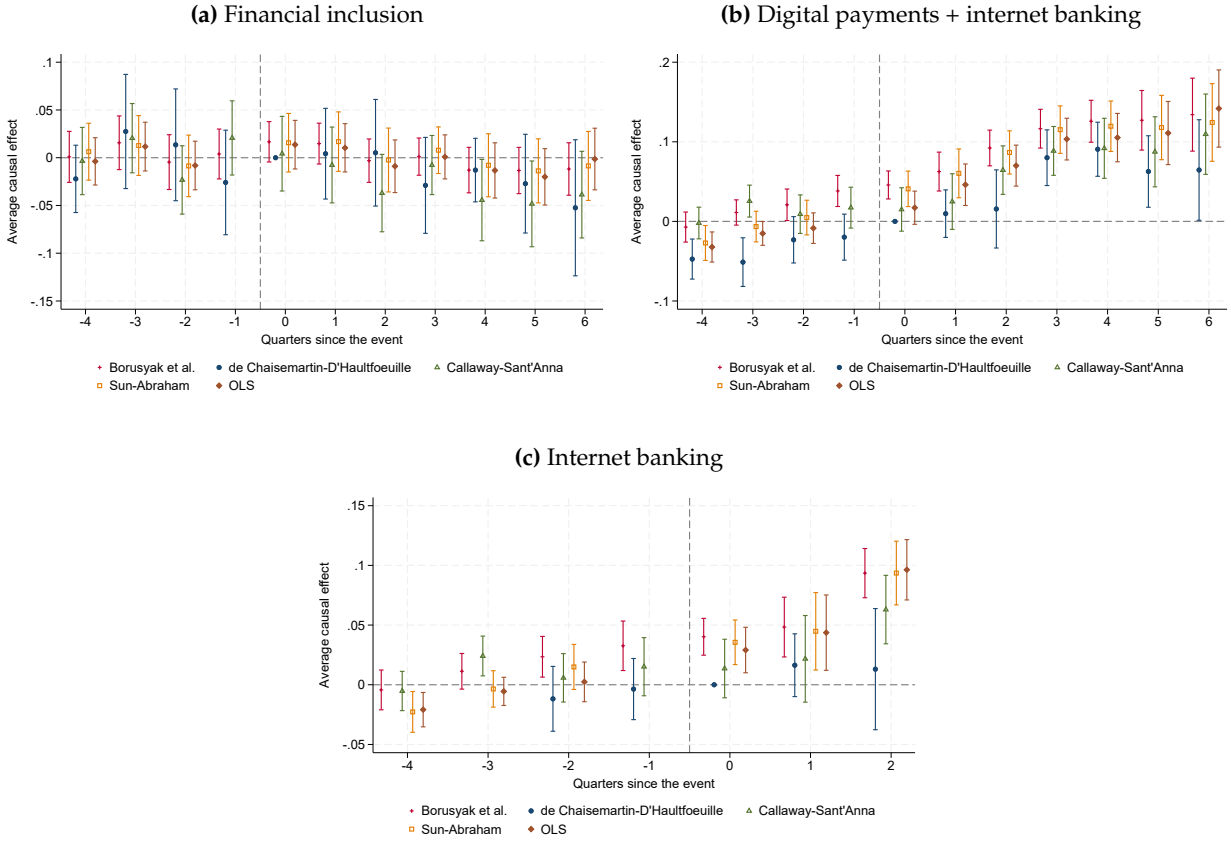
Note: This figure reports dynamic effects of district-level exposure to interoperability on card use (debit and credit) by expenditure category. The exposure measure is constructed from district-level deposits in the pre-interoperability period (December 2022) for institutions participating in each phase of the strategy. Outcomes are indicators equal to one if the individual reports using at least one type of card for the corresponding expenditure category. Estimation uses staggered-adoption difference-in-differences methods following Callaway and Sant'Anna (2021); de Chaisemartin and D'Haultfœuille (2024); Borusyak et al. (2024); Sun and Abraham (2021). Standard errors are clustered at the district level.

Figure A.9: Dynamic effects on card use by expenditure category



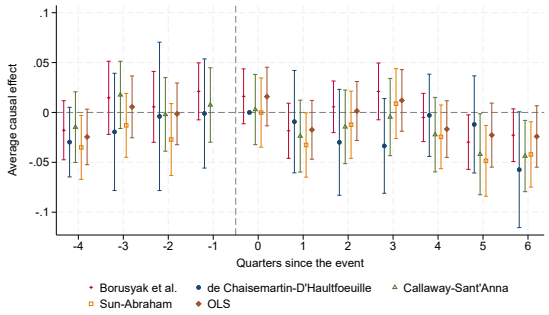
Note: This figure reports dynamic effects of district-level exposure to interoperability on card use (debit and credit) by expenditure category. The exposure measure is constructed from district-level deposits in the pre-interoperability period (December 2022) for institutions participating in each phase of the strategy. Outcomes are indicators equal to one if the individual reports using at least one type of card for the corresponding expenditure category. Estimation uses staggered-adoption difference-in-differences methods following Callaway and Sant'Anna (2021); de Chaisemartin and D'Haultfœuille (2024); Borusyak et al. (2024); Sun and Abraham (2021). Standard errors are clustered at the district level.

Figure A.10: Dynamic effects on financial inclusion and digital payment use



Note: This figure reports dynamic effects of district-level exposure to interoperability on financial inclusion, digital payment use, and internet banking. The exposure measure is constructed from district-level deposits in the pre-interoperability period (December 2022) for institutions participating in each phase of the strategy. For financial inclusion, the outcome is an indicator equal to one if the individual reports holding any type of bank account. For internet banking, the outcome is an indicator equal to one if the individual reports using internet banking for at least one of the nine expenditure categories. The variable IB + Digital Wallets additionally incorporates digital wallet use in 2024. Estimation uses staggered-adoption difference-in-differences methods following Callaway and Sant'Anna (2021); de Chaisemartin and D'Haultfœuille (2024); Borusyak et al. (2024); Sun and Abraham (2021). Standard errors are clustered at the district level.

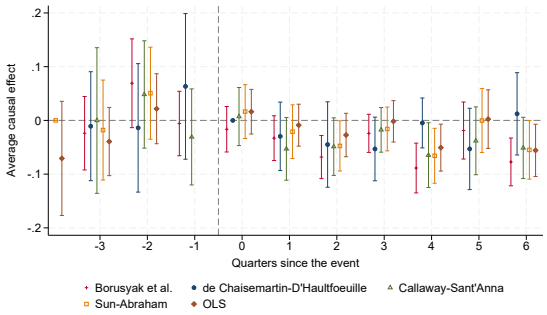
Figure A.11: Dynamic effects on household financial fragility



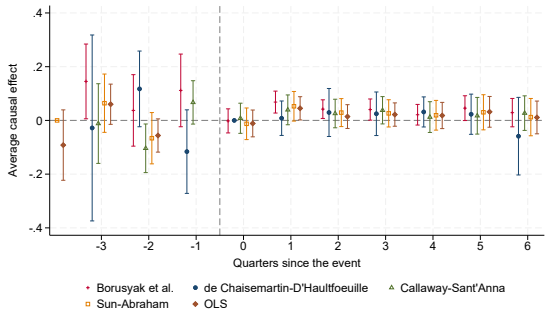
Note: This figure reports dynamic effects of district-level exposure to interoperability on a measure of household financial fragility. The exposure measure is constructed from district-level deposits in the pre-interoperability period (December 2022) for institutions participating in each phase of the strategy. Financial fragility is an indicator equal to one if the individual reports that, given the household's current economic situation, the household is forced to draw down savings or borrow. Estimation uses staggered-adoption difference-in-differences methods following Callaway and Sant'Anna (2021); de Chaisemartin and D'Haultfœuille (2024); Borusyak et al. (2024); Sun and Abraham (2021). Standard errors are clustered at the district level.

Figure A.12: Mechanisms: dynamic effects on internet access

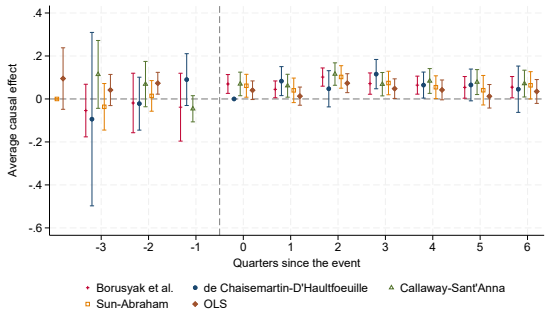
(a) Fixed



(b) Postpaid mobile



(c) Prepaid mobile



Note: This figure reports dynamic effects of district-level exposure to interoperability on three measures of household internet access. The exposure measure is constructed from district-level deposits in the pre-interoperability period (December 2022) for institutions participating in each phase of the strategy. Outcomes are indicators equal to one if the household reports having access to fixed internet, postpaid mobile internet, and prepaid mobile internet, respectively. Estimation uses staggered-adoption difference-in-differences methods following Callaway and Sant'Anna (2021); de Chaisemartin and D'Haultfœuille (2024); Borusyak et al. (2024); Sun and Abraham (2021). Standard errors are clustered at the district level.

Table A.2: Robustness to the expanded exposure measure (deposits + ATMs/agents): effect on cash use

	Interoperability Exposure x Post	Std. Error	p-value	p BH	Pre-treatment mean
Groceries	-0.015	0.009	0.082	0.140	0.911
Ready-to-eat food	-0.013	0.008	0.084	0.140	0.909
Laundry	-0.014	0.009	0.102	0.145	0.896
Utilities	-0.030**	0.011	0.004	0.022	0.859
Cooking fuels	-0.046***	0.013	0.000	0.004	0.794
Personal hygiene	-0.010	0.007	0.146	0.162	0.910
Clothing and footwear	-0.025**	0.010	0.011	0.036	0.838
Furniture and appliances	-0.033	0.021	0.116	0.145	0.253
Household appliances	-0.028*	0.013	0.031	0.077	0.106
Other	-0.003	0.014	0.851	0.851	0.836
N			233,196		
N Clusters			1,347		

Note: This table reports the effect of district-level exposure to interoperability on cash use by expenditure category. The exposure measure is constructed using district-level deposits and the number of ATMs and banking agents, measured prior to interoperability (December 2022), for institutions participating in each phase of the strategy. Outcomes are indicators equal to one if the individual reports using cash for the corresponding expenditure category. Estimation uses the staggered-adoption difference-in-differences method following Callaway and Sant'Anna (2021). Standard errors are clustered at the district level. Statistical significance indicated by *, **, and *** is based on the adjusted p-value reported in column *p* BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the mean of each outcome in the pre-treatment period (before 2023Q2).

Table A.3: Robustness to the expanded exposure measure (deposits + ATMs/agents): effect on card use

	Interoperability Exposure x Post	Std. Error	p-value	p BH	Pre-treatment mean
Groceries	-0.013*	0.006	0.032	0.054	0.055
Ready-to-eat food	-0.009	0.006	0.126	0.180	0.050
Laundry	-0.013**	0.006	0.022	0.044	0.047
Utilities	-0.014***	0.004	0.000	0.004	0.031
Cooking fuels	-0.011***	0.003	0.001	0.007	0.019
Personal hygiene	-0.013**	0.005	0.014	0.044	0.048
Clothing and footwear	-0.013**	0.006	0.022	0.044	0.054
Furniture and appliances	-0.004	0.003	0.167	0.208	0.012
Household appliances	-0.001	0.003	0.711	0.711	0.012
Other	-0.002	0.004	0.591	0.656	0.023
N			233,196		
N Clusters			1,347		

Note: This table reports the effect of district-level exposure to interoperability on card use (debit and credit) by expenditure category. The exposure measure is constructed using district-level deposits and the number of ATMs and banking agents, measured prior to interoperability (December 2022), for institutions participating in each phase of the strategy. Outcomes are indicators equal to one if the individual reports using at least one type of card for the corresponding expenditure category. Estimation uses the staggered-adoption difference-in-differences method following Callaway and Sant'Anna (2021). Standard errors are clustered at the district level. Statistical significance indicated by *, **, and *** is based on the adjusted p-value reported in column p BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the mean of each outcome in the pre-treatment period (before 2023Q2).

Table A.4: Robustness to the expanded exposure measure (deposits + ATMs/agents): effect on digital payments and financial inclusion

	Interoperability Exposure x Post	Std. Error	p-value	p BH	Pre-treatment mean	N	Clusters
Financial Inclusion	-0.019	0.016	0.237	0.237	0.524	233,196	1,347
Internet Banking (IB)	0.034***	0.012	0.005	0.008	0.060	155,995	1,306
IB + Digital Wallets	0.033***	0.011	0.003	0.010	0.060	233,196	1,347

Note: This table reports the effect of district-level exposure to interoperability on financial inclusion, internet banking use, and digital payments. For financial inclusion, the outcome is an indicator equal to one if the individual reports holding any type of bank account. For internet banking, the outcome is an indicator equal to one if the individual reports using internet banking for at least one of the nine expenditure categories. The variable IB + Digital Wallets additionally incorporates digital wallet use in 2024. The exposure measure is constructed using district-level deposits and the number of ATMs and banking agents, measured prior to interoperability (December 2022), for institutions participating in each phase of the strategy. Estimation uses the staggered-adoption difference-in-differences method following Callaway and Sant'Anna (2021). Starting in 2024, INEI includes mobile banking and ATMs as options in the internet-banking question, so I restrict the sample through 2023 for the internet-banking outcome. I also report estimates for IB + Digital Wallets, which leverages the expanded set of questions in 2024 on mobile banking, ATMs, and digital wallets. Standard errors are clustered at the district level. Statistical significance indicated by *, **, and *** is based on the adjusted p-value reported in column p BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the mean of each outcome in the pre-treatment period (before 2023Q2).

Table A.5: Robustness using March 2023 deposits for exposure: effect on cash use

	Interoperability Exposure x Post	Std. Error	p-value	p BH	Pre-treatment mean
Groceries	-0.017*	0.009	0.055	0.069	0.911
Ready-to-eat food	-0.019**	0.008	0.023	0.047	0.909
Laundry	-0.017*	0.009	0.048	0.069	0.896
Utilities	-0.044***	0.011	0.000	0.001	0.859
Cooking fuels	-0.051***	0.013	0.000	0.001	0.794
Personal hygiene	-0.015*	0.007	0.038	0.064	0.910
Clothing and footwear	-0.031***	0.010	0.002	0.008	0.838
Furniture and appliances	-0.038*	0.021	0.073	0.081	0.253
Household appliances	-0.031**	0.013	0.016	0.041	0.106
Other	-0.005	0.014	0.700	0.700	0.836
N			233,196		
N Clusters			1,347		

Note: This table reports the effect of district-level exposure to interoperability on cash use by expenditure category. The exposure measure is constructed using district-level deposits measured prior to interoperability (March 2023) for institutions participating in each phase of the strategy. Outcomes are indicators equal to one if the individual reports using cash for the corresponding expenditure category. Estimation uses the staggered-adoption difference-in-differences method following Callaway and Sant'Anna (2021). Standard errors are clustered at the district level. Statistical significance indicated by *, **, and *** is based on the adjusted p-value reported in column p BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the mean of each outcome in the pre-treatment period (before 2023Q2).

Table A.6: Robustness using March 2023 deposits for exposure: effect on card use

	Interoperability Exposure x Post	Std. Error	p-value	p BH	Pre-treatment mean
Groceries	-0.014**	0.006	0.021	0.041	0.055
Ready-to-eat food	-0.010	0.006	0.091	0.130	0.050
Laundry	-0.014**	0.006	0.019	0.041	0.047
Utilities	-0.017***	0.004	0.000	0.001	0.031
Cooking fuels	-0.010**	0.003	0.002	0.011	0.019
Personal hygiene	-0.017**	0.006	0.003	0.011	0.048
Clothing and footwear	-0.013**	0.006	0.028	0.046	0.054
Furniture and appliances	-0.003	0.003	0.410	0.513	0.012
Household appliances	0.001	0.003	0.680	0.756	0.012
Other	0.001	0.004	0.839	0.839	0.023
N			233,196		
N Clusters			1,347		

Note: This table reports the effect of district-level exposure to interoperability on card use (debit and credit) by expenditure category. The exposure measure is constructed using district-level deposits measured prior to interoperability (March 2023) for institutions participating in each phase of the strategy. Outcomes are indicators equal to one if the individual reports using at least one type of card for the corresponding expenditure category. Estimation uses the staggered-adoption difference-in-differences method following Callaway and Sant'Anna (2021). Standard errors are clustered at the district level. Statistical significance indicated by *, **, and *** is based on the adjusted p-value reported in column p BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the mean of each outcome in the pre-treatment period (before 2023Q2).

Table A.7: Robustness using March 2023 deposits for exposure: effect on digital payments and financial inclusion

	Interoperability Exposure x Post	Std. Error	p-value	p BH	Pre-treatment mean	N	N Clusters
Financial Inclusion	-0.025	0.016	0.125	0.125	0.524	233,196	1,347
Internet Banking (IB)	0.033***	0.012	0.006	0.009	0.060	155,995	1,306
IB + Digital Wallets	0.051***	0.011	0.000	0.000	0.060	233,196	1,347

Note: This table reports the effect of district-level exposure to interoperability on financial inclusion, internet banking use, and digital payments. For financial inclusion, the outcome is an indicator equal to one if the individual reports holding any type of bank account. For internet banking, the outcome is an indicator equal to one if the individual reports using internet banking for at least one of the nine expenditure categories. The variable IB + Digital Wallets additionally incorporates digital wallet use in 2024. The exposure measure is constructed using district-level deposits measured prior to interoperability (March 2023) for institutions participating in each phase of the strategy. Estimation uses the staggered-adoption difference-in-differences method following Callaway and Sant'Anna (2021). Starting in 2024, INEI includes mobile banking and ATMs as options in the internet-banking question, so I restrict the sample through 2023 for the internet-banking outcome. I also report estimates for IB + Digital Wallets, which leverages the expanded set of questions in 2024 on mobile banking, ATMs, and digital wallets. Standard errors are clustered at the district level. Statistical significance indicated by *, **, and *** is based on the adjusted p-value reported in column p BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the mean of each outcome in the pre-treatment period (before 2023Q2).

Table A.8: Robustness restricting to districts with at least one ATM or banking agent: effect on cash use

	Interoperability Exposure x Post	Std. Error	p-value	p BH	Pre-treatment mean
Groceries	-0.016*	0.009	0.072	0.090	0.910
Ready-to-eat food	-0.020**	0.008	0.019	0.039	0.909
Laundry	-0.017*	0.009	0.047	0.078	0.896
Utilities	-0.044***	0.011	0.000	0.001	0.858
Cooking fuels	-0.052***	0.013	0.000	0.001	0.794
Personal hygiene	-0.014*	0.007	0.056	0.080	0.910
Clothing and footwear	-0.031**	0.010	0.003	0.010	0.837
Furniture and appliances	-0.037*	0.021	0.086	0.095	0.251
Household appliances	-0.030**	0.013	0.019	0.039	0.106
Other	-0.005	0.014	0.734	0.734	0.835
N			226,304		
N Clusters			1,252		

Note: This table reports the effect of district-level exposure to interoperability on cash use by expenditure category. The exposure measure is constructed using district-level deposits measured prior to interoperability (December 2022) for institutions participating in each phase of the strategy. Outcomes are indicators equal to one if the individual reports using cash for the corresponding expenditure category. Estimation restricts the sample to districts with at least one ATM or banking agent and uses the staggered-adoption difference-in-differences method following Callaway and Sant'Anna (2021). Standard errors are clustered at the district level. Statistical significance indicated by *, **, and *** is based on the adjusted p-value reported in column p BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the mean of each outcome in the pre-treatment period (before 2023Q2).

Table A.9: Robustness restricting to districts with at least one ATM or banking agent: effect on card use

	Interoperability Exposure x Post	Std. Error	p-value	p BH	Pre-treatment mean
Groceries	-0.015**	0.006	0.017	0.030	0.055
Ready-to-eat food	-0.010	0.006	0.088	0.126	0.051
Laundry	-0.014**	0.006	0.014	0.030	0.048
Utilities	-0.017***	0.004	0.000	0.000	0.031
Cooking fuels	-0.011***	0.003	0.001	0.007	0.019
Personal hygiene	-0.017***	0.006	0.003	0.008	0.048
Clothing and footwear	-0.014**	0.006	0.018	0.030	0.055
Furniture and appliances	-0.003	0.003	0.388	0.486	0.012
Household appliances	0.001	0.003	0.754	0.838	0.012
Other	0.001	0.004	0.889	0.889	0.023
N			226,304		
N Clusters			1,252		

Note: This table reports the effect of district-level exposure to interoperability on card use (debit and credit) by expenditure category. The exposure measure is constructed using district-level deposits measured prior to interoperability (December 2022) for institutions participating in each phase of the strategy. Outcomes are indicators equal to one if the individual reports using at least one type of card for the corresponding expenditure category. Estimation restricts the sample to districts with at least one ATM or banking agent and uses the staggered-adoption difference-in-differences method following Callaway and Sant'Anna (2021). Standard errors are clustered at the district level. Statistical significance indicated by *, **, and *** is based on the adjusted p-value reported in column p BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the mean of each outcome in the pre-treatment period (before 2023Q2).

Table A.10: Robustness restricting to districts with at least one ATM or banking agent: effect on digital payments and financial inclusion

	Interoperability Exposure x Post	Std. Error	p-value	p BH	Pre-treatment mean	N	Clusters
Financial Inclusion	-0.024	0.017	0.140	0.140	0.525	226,304	1,252
Internet Banking (IB)	0.033**	0.012	0.007	0.011	0.061	151,356	1,218
IB + Digital Wallets	0.049***	0.011	0.000	0.000	0.061	226,304	1,252

Note: This table reports the effect of district-level exposure to interoperability on financial inclusion, internet banking use, and digital payments. For financial inclusion, the outcome is an indicator equal to one if the individual reports holding any type of bank account. For internet banking, the outcome is an indicator equal to one if the individual reports using internet banking for at least one of the nine expenditure categories. The variable IB + Digital Wallets additionally incorporates digital wallet use in 2024. The exposure measure is constructed using district-level deposits measured prior to interoperability (December 2022) for institutions participating in each phase of the strategy. Estimation restricts the sample to districts with at least one ATM or banking agent and uses the staggered-adoption difference-in-differences method following Callaway and Sant'Anna (2021). Starting in 2024, INEI includes mobile banking and ATMs as options in the internet-banking question, so I restrict the sample through 2023 for the internet-banking outcome. I also report estimates for IB + Digital Wallets, which leverages the expanded set of questions in 2024 on mobile banking, ATMs, and digital wallets. Standard errors are clustered at the district level. Statistical significance indicated by *, **, and *** is based on the adjusted p-value reported in column p BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the mean of each outcome in the pre-treatment period (before 2023Q2).

Table A.11: Placebo: effect on cash use

	Interoperability Exposure x Post	Std. Error	p-value	p BH	Pre-treatment mean
Groceries	0.003	0.008	0.724	0.994	0.911
Ready-to-eat food	-0.002	0.010	0.807	0.994	0.905
Laundry	-0.002	0.009	0.777	0.994	0.897
Utilities	-0.001	0.011	0.924	0.994	0.858
Cooking fuels	0.011	0.014	0.440	0.994	0.795
Personal hygiene	0.001	0.009	0.889	0.994	0.910
Clothing and footwear	0.020	0.014	0.161	0.538	0.828
Furniture and appliances	-0.000	0.022	0.994	0.994	0.250
Household appliances	0.020	0.014	0.144	0.538	0.104
Other	0.058**	0.020	0.003	0.032	0.826
N			136,080		
N Clusters			1,276		

Note: This table reports a placebo exercise to assess the validity of the identification strategy. I restrict the sample to the pre-treatmen period 2021Q3–2023Q1 and assign placebo treatment dates to the three cohorts, imputing adoption dates 2022Q1, 2022Q2, and 2022Q3 for the cohorts that in the actual rollout begin in 2023Q2, 2023Q4, and 2024Q1, respectively. I keep the same predetermined exposure measure constructed from district deposits (December 2022) of participating institutions in each phase and estimate a staggered-adoption difference-in-differences model following Callaway and Sant'Anna (2021), clustering standard errors at the district level. Outcomes are indicators equal to one if the individual reports using cash in each expenditure category. Statistical significance indicated by *, **, and *** is based on the adjusted p-value reported in column p BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the outcome mean in the pre-treatment period (before 2023Q2).

Table A.12: Placebo: effect on card use

	Interoperability Exposure x Post	Std. Error	p-value	p BH	Pre-treatment mean
Groceries	-0.007	0.007	0.265	0.520	0.053
Ready-to-eat food	-0.008	0.007	0.233	0.520	0.049
Laundry	-0.007	0.007	0.312	0.520	0.046
Utilities	-0.000	0.004	0.934	0.934	0.029
Cooking fuels	0.006	0.003	0.021	0.205	0.017
Personal hygiene	-0.003	0.006	0.626	0.783	0.046
Clothing and footwear	-0.002	0.006	0.715	0.795	0.052
Furniture and appliances	0.004	0.003	0.113	0.520	0.011
Household appliances	-0.003	0.003	0.289	0.520	0.012
Other	0.002	0.004	0.549	0.783	0.022
N			136,080		
N Clusters			1,276		

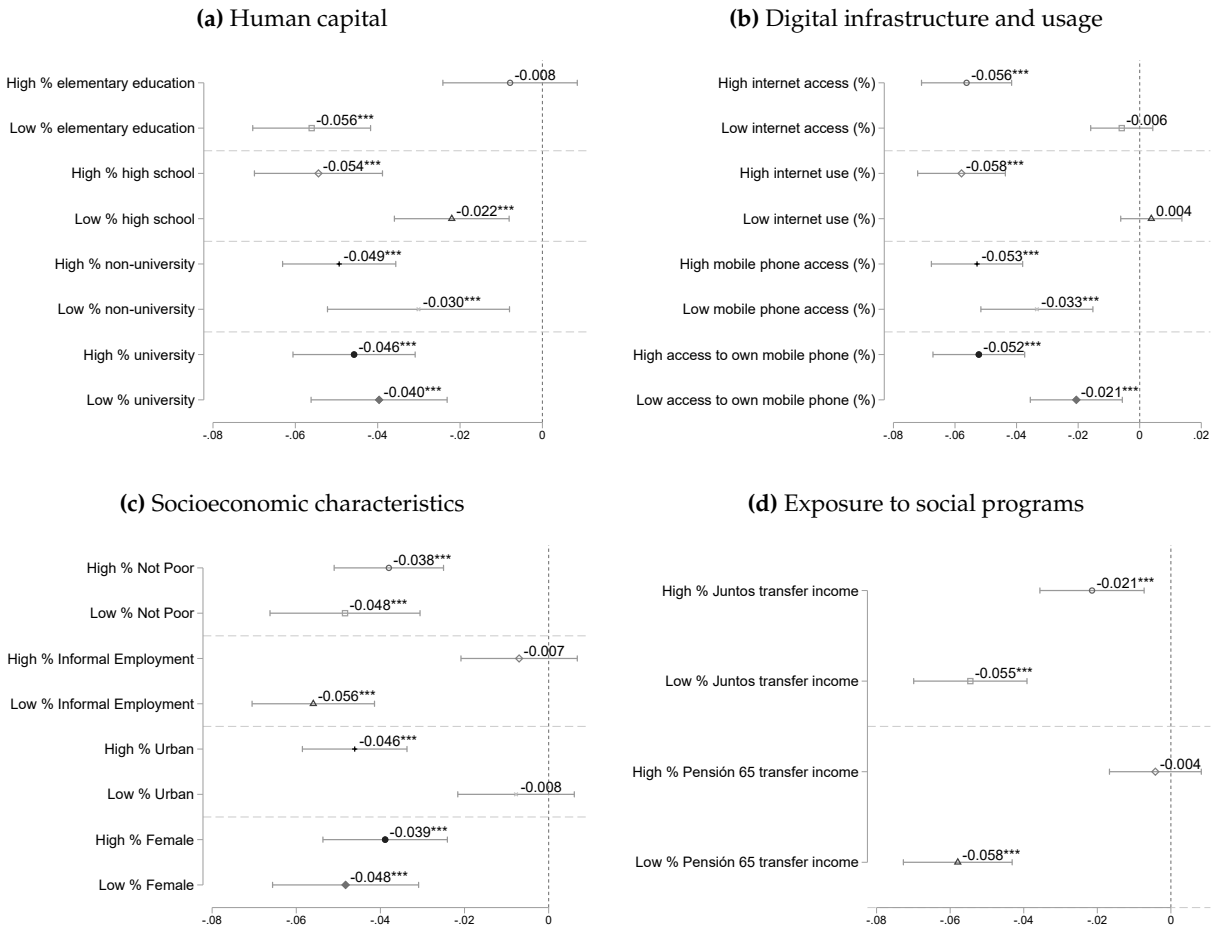
Note: This table reports a placebo exercise to assess the validity of the identification strategy. I restrict the sample to the pre-treatmen period 2021Q3–2023Q1 and assign placebo treatment dates to the three cohorts, imputing adoption dates 2022Q1, 2022Q2, and 2022Q3 for the cohorts that in the actual rollout begin in 2023Q2, 2023Q4, and 2024Q1, respectively. I keep the same predetermined exposure measure constructed from district deposits (December 2022) of participating institutions in each phase and estimate a staggered-adoption difference-in-differences model following Callaway and Sant'Anna (2021), clustering standard errors at the district level. Outcomes are indicators equal to one if the individual reports using a card (debit or credit) in each expenditure category. Statistical significance indicated by *, **, and *** is based on the adjusted p-value reported in column p BH, computed using the Benjamini–Hochberg procedure to correct for multiple testing (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The last column reports the outcome mean in the pre-treatment period (before 2023Q2).

Table A.13: Placebo: effect on digital payments and financial inclusion

	Interoperability Exposure x Post	Std. Error	p-value	p BH	Pre-treatment mean	N	Clusters
Financial Inclusion	0.001	0.014	0.962	0.962	0.520	136,080	1,276
Internet Banking (IB)	0.009	0.007	0.148	0.445	0.056	136,080	1,276
IB + Digital Wallets	0.009	0.007	0.148	0.223	0.056	136,080	1,276

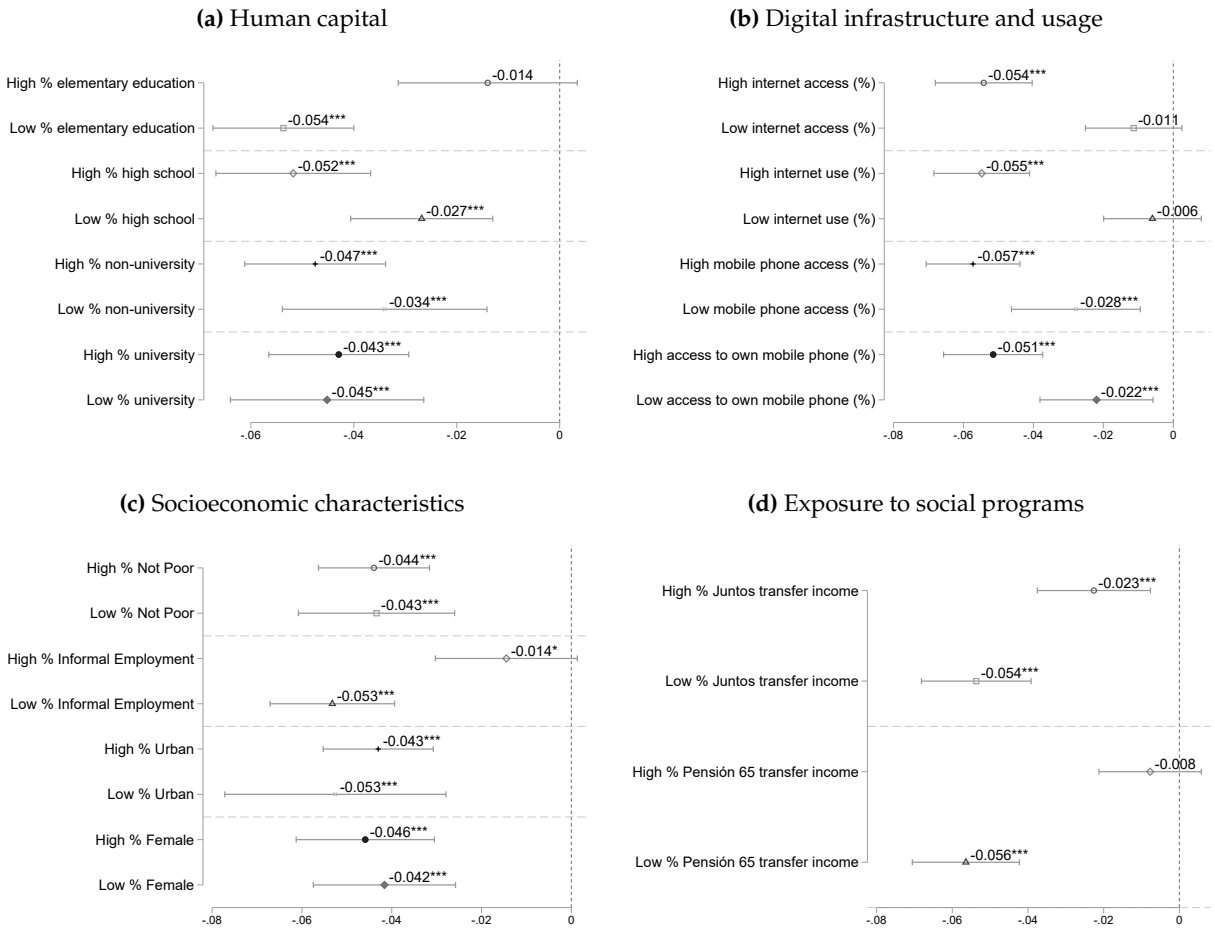
Note: This table reports a placebo exercise restricting the sample to the pre-treatmen period 2021Q3–2023Q1 and imputing placebo adoption dates (2022Q1, 2022Q2, and 2022Q3) to the three cohorts. I keep the same predetermined exposure measure (district deposits in December 2022 of participating institutions) and estimate a staggered-adoption difference-in-differences model following Callaway and Sant'Anna (2021), with standard errors clustered at the district level. Outcomes include individual indicators for financial inclusion (having an account), internet banking (IB), and a harmonized measure IB + Digital Wallets; because the INEI question changes starting in 2024, estimation for IB is restricted through 2023. Statistical significance is based on Benjamini–Hochberg adjusted p-values (column p BH), and the last column reports the pre-treatment mean (before 2023Q2). (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$).

Figure A.13: Heterogeneous effects on cash use for utilities



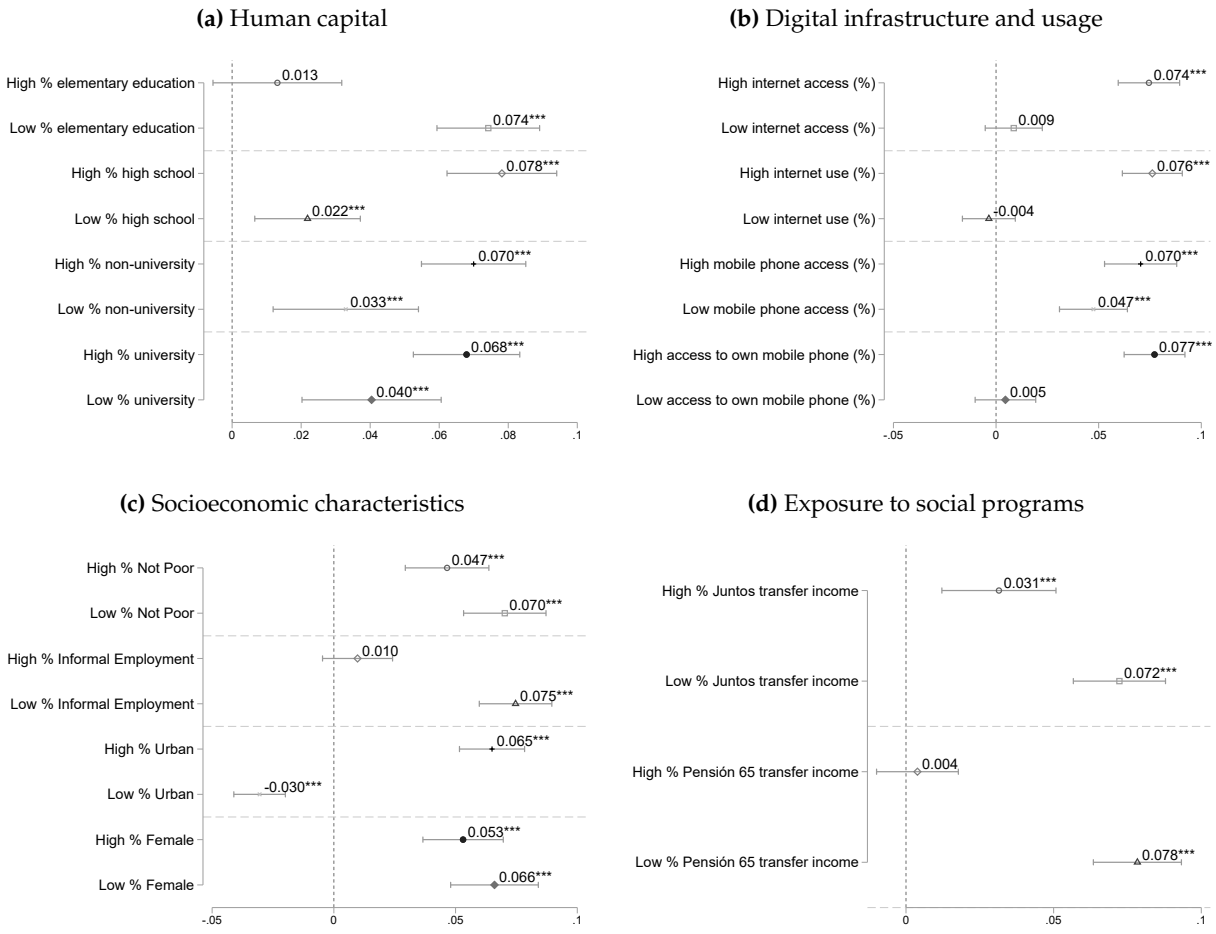
Note: This figure reports heterogeneous-effects estimates of interoperability on cash use for utilities, using the imputation-based difference-in-differences estimator of Borusyak et al. (2024) with standard errors clustered at the district level. The outcome is an indicator equal to one if the individual reports using cash for housing payments. Heterogeneity is defined using indicators for district pre-treatment characteristics measured in 2022, splitting districts into "high" and "low" groups depending on whether each covariate is above its median. The panels summarize four dimensions: human capital, digital infrastructure and usage, socioeconomic characteristics, and exposure to social programs. Coefficients are interpreted as percentage-point changes.

Figure A.14: Heterogeneous effects on cash use for cooking fuels



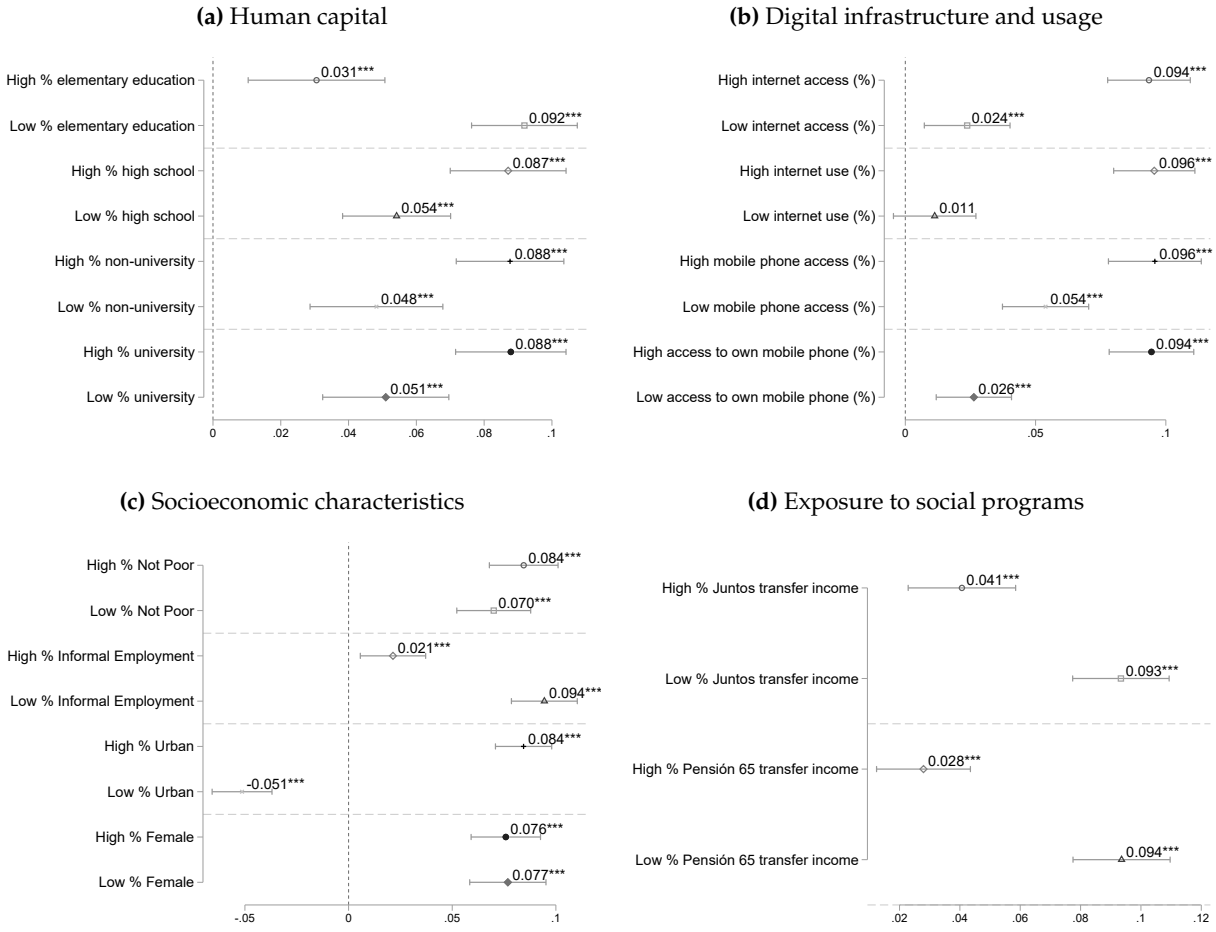
Note: This figure reports heterogeneous-effects estimates of interoperability on cash use for cooking fuels, using the imputation-based difference-in-differences estimator of Borusyak et al. (2024) with standard errors clustered at the district level. The outcome is an indicator equal to one if the individual reports using cash for cooking-fuel expenditures. Heterogeneity is defined using indicators for district pre-treatment characteristics measured in 2022, splitting districts into "high" and "low" groups depending on whether each covariate is above its median. The panels summarize four dimensions: human capital, digital infrastructure and usage, socioeconomic characteristics, and exposure to social programs. Coefficients are interpreted as percentage-point changes.

Figure A.15: Heterogeneous effects on internet banking use



Note: This figure reports heterogeneous-effects estimates of interoperability on internet banking use, using the imputation-based difference-in-differences estimator of Borusyak et al. (2024) with standard errors clustered at the district level. The outcome here is an indicator equal to one if the individual reports using internet banking (not cash for housing payments). Heterogeneity is defined using indicators for district pre-treatment characteristics measured in 2022, splitting districts into "high" and "low" groups depending on whether each covariate is above its median. The panels summarize four dimensions: human capital, digital infrastructure and usage, socioeconomic characteristics, and exposure to social programs. Coefficients are interpreted as percentage-point changes.

Figure A.16: Heterogeneous effects on digital payment use



Note: This figure reports heterogeneous-effects estimates of interoperability on digital payment use, using the imputation-based difference-in-differences estimator of Borusyak et al. (2024) with standard errors clustered at the district level. The outcome here is an indicator equal to one if the individual reports using digital payments (internet banking and/or digital wallets, depending on availability). Heterogeneity is defined using indicators for district pre-treatment characteristics measured in 2022, splitting districts into "high" and "low" groups depending on whether each covariate is above its median. The panels summarize four dimensions: human capital, digital infrastructure and usage, socioeconomic characteristics, and exposure to social programs. Coefficients are interpreted as percentage-point changes.